PHYSICS GRADE 10: INTRODUCTION TO ELECTRONICS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.3 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 3.0: Electricity and Magnetism
Sub-Strand	Sub-Strand 3.3: Introduction to Electronics
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Conductors, insulators, and semiconductors — classification and properties – Energy band theory — valence band, conduction band, and band gap – Intrinsic semiconductors — pure silicon and germanium; electron-hole pair generation – Extrinsic semiconductors — n-type doping (phosphorus/arsenic donors) and p-type doping (boron/aluminium acceptors) – The p-n junction — formation, depletion region, forward and reverse bias – Superconductors — zero electrical resistance, critical temperature, and the Meissner effect – Applications of semiconductors and superconductors — solar panels (photovoltaic cells), microchips/integrated circuits, superconducting magnets in MRI machines and maglev trains
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) distinguish between conductors, insulators, and semiconductors using energy band theory, b) explain the electrical behaviour of intrinsic semiconductors in terms of electron-hole pair generation, c) describe how doping produces n-type and p-type extrinsic semiconductors, d) explain the formation and behaviour of a p-n junction under forward and reverse bias conditions, e) describe the properties of superconductors and the significance of critical temperature, f) analyse real-world applications of semiconductors and superconductors including solar panels, microchips, and superconducting magnets, g) appreciate the role of electronics in driving technological advancement and solving societal challenges in Kenya and beyond.
Core Competencies	– Communication and Collaboration: learners work in groups during the PhET semiconductor simulation (Lesson 5) and case-study analysis (Lesson 7), sharing findings, negotiating explanations, and presenting conclusions to the class — building teamwork and clear scientific communication. – Critical Thinking and Problem Solving: learners analyse why a seemingly non-conducting material (silicon) can be engineered to conduct electricity, construct evidence-based arguments using energy band diagrams, and evaluate trade-offs in semiconductor device design. – Learning to Learn: learners revise and extend their mental models across the lesson sequence — from a basic conductor/insulator binary to a nuanced understanding of doping, junctions, and superconductivity — practising metacognitive reflection on how their thinking has changed. – Digital Literacy: learners interact with the PhET "Semiconductor" simulation and access the Energy Band Theory YouTube

	recap video, building confidence in using digital tools for scientific inquiry. – Creativity and Imagination: learners design annotated conceptual models of p-n junctions and propose how semiconductor or superconductor technology could address a specific Kenyan challenge (e.g., affordable solar energy in off-grid Turkana County).
Core Values	– Responsibility: learners take ownership of group roles during simulation activities and case-study investigations, ensuring every member contributes meaningfully to the team's findings. – Respect: learners demonstrate open-mindedness when peers propose alternative explanations for semiconductor behaviour, appreciating diverse reasoning before reaching consensus. – Unity: learners collaborate across different backgrounds and abilities during group model-building, recognising that collective reasoning produces stronger scientific explanations than individual effort alone. – Social Justice: learners reflect on equitable access to semiconductor technology (e.g., affordable solar panels for rural Kenyan communities) and consider the ethical responsibilities of engineers and policymakers.
Science & Engineering Practices	– SEP 1 — Asking Questions and Defining Problems: learners generate testable questions about why certain materials conduct electricity under some conditions but not others, anchored to the puzzling behaviour of silicon in solar panels observed in the phenomenon. – SEP 2 — Developing and Using Models: learners construct and iteratively revise energy band diagrams and p-n junction models across Lessons 1–6, culminating in a cumulative annotated model that explains the anchoring phenomenon. – SEP 3 — Planning and Carrying Out Investigations: learners design and execute the PhET semiconductor simulation in Lesson 5, varying doping concentration and bias voltage and recording systematic observations. – SEP 4 — Analysing and Interpreting Data: learners analyse I–V characteristic graphs for p-n junctions and interpret simulation data to identify patterns linking doping type/level to conductivity. – SEP 6 — Constructing Explanations and Designing Solutions: learners construct evidence-based explanations for how a photovoltaic solar cell converts sunlight to electricity, connecting band theory, the p-n junction, and electron flow. – SEP 8 — Obtaining, Evaluating, and Communicating Information: learners evaluate case-study sources on solar panels, microchips, and superconducting magnets in Lesson 7 and communicate findings through structured presentations.
Pertinent & Contemporary Issues (PCIs)	– Environmental Education: learners connect semiconductor photovoltaic technology to Kenya's renewable energy goals — reducing reliance on fossil fuels and mitigating climate change impacts felt acutely in regions like the Rift Valley and Turkana Basin. – Gender Disparity: learners challenge stereotypes that Electronics and Physics are male-dominated fields by profiling Kenyan women working in technology and engineering sectors (e.g., women engineers at Kenya Power, female researchers at JKUAT), encouraging all learners to see themselves as future electronics professionals. – Financial Literacy: learners evaluate the economic cost-benefit of semiconductor-based solar panels for rural Kenyan households versus grid electricity, developing awareness of technology as a driver of economic empowerment. – Citizenship: learners discuss Kenya's Silicon Savannah (Nairobi's tech hub) and consider how a citizenry that understands electronics can drive local innovation and reduce dependence on imported technology.
Career Connections	– Electrical/Electronics Engineer: this sub-strand provides foundational understanding of semiconductors and circuit components essential for designing consumer electronics, power systems, and communication devices manufactured or used across Kenya. – Solar Energy Technician/Renewable Energy Engineer: understanding p-n junctions and photovoltaic cells directly underpins careers in Kenya's rapidly growing solar energy sector, including installation and design of off-grid solar systems in rural counties. – Medical Physicist/Biomedical Engineer: knowledge of superconducting magnets connects to careers operating and

	maintaining MRI machines in Kenyan hospitals such as Kenyatta National Hospital and Aga Khan University Hospital. – Computer Hardware Engineer/ICT Specialist: understanding how doped semiconductors form transistors and integrated circuits is foundational for careers in Kenya's Silicon Savannah tech ecosystem, including chip design and embedded systems. – Research Scientist (Materials Science/Nanotechnology): study of energy bands and superconductivity opens pathways to postgraduate research at Kenyan universities (University of Nairobi, JKUAT, Maseno) focusing on advanced materials for energy and health applications.
Focus for Lessons	This unit introduces Grade 10 learners to the physics of electronic materials, starting from the fundamental question of why some materials conduct electricity and others do not. Using energy band theory as a conceptual backbone, learners progress from pure (intrinsic) semiconductors through engineered (extrinsic) doped materials to the remarkable world of superconductors, always connecting abstract models to tangible Kenyan applications. The instructional approach is phenomenon-driven and model-centred: learners repeatedly revise a cumulative diagram as their understanding deepens, grounding every concept in the puzzling real-world behaviour of the silicon solar panel introduced in Lesson 1.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun than on a cool morning — and what is so special about silicon that it can act as both a conductor and an insulator depending on the conditions? KICD KEY INQUIRY QUESTIONS: 1. How do the electrical properties of materials depend on their atomic and energy band structure? 2. How has the understanding and engineering of semiconductor materials transformed modern technology and everyday life?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Solar Panel That Works Better on a Cool, Cloudy Day in Nairobi" A farmer in Kajiado County installs a silicon solar panel to power a water pump for irrigation. On a hot, cloudless afternoon (≈35 °C), the panel generates noticeably less electrical power than on a cooler, overcast morning (≈18 °C). This is deeply puzzling: learners expect more sunlight = more electricity, yet the data show the opposite trend for power output efficiency. Furthermore, the panel produces no electricity at all in complete darkness, yet silicon — the material the panel is made from — is neither a metal conductor nor a plastic insulator. How can a single material sometimes conduct electricity and sometimes not? And why does heating it up make it work less efficiently rather than more, as would happen with a simple resistor? This phenomenon cannot be explained by anything learners have studied before about simple circuits and Ohm's Law — it demands a new model of how electrons behave inside solid materials.
Storyline Thread	Lesson 1–2 (Conductors, Insulators, Semiconductors): Learners encounter the anchoring phenomenon — the Kajiado solar panel puzzle. They sort familiar Kenyan materials (copper wire, plastic mkono wa sufuria handle, graphite pencil, silicon chip) into conductors/insulators by prediction, then test using a simple circuit. The unexpected in-between behaviour of silicon and graphite creates productive confusion. Learners are introduced to energy band theory as the new model needed to explain what they observe, constructing initial band diagrams for a metal, an insulator, and a semiconductor. The Driving Question Board is opened. Lesson 3 (Intrinsic Semiconductors): Learners deepen the band model by exploring pure (intrinsic) silicon — how thermal energy at room temperature excites a small number of electrons across the band gap, creating electron-hole pairs. They connect this to why the solar panel works at all without doping. A temperature–conductivity graph for pure silicon is analysed, beginning to explain why heat affects semiconductor behaviour differently from metals. Lesson 4 (Extrinsic Semiconductors — n-type & p-type): Learners investigate how adding tiny amounts of impurity atoms (doping) dramatically increases conductivity without changing the crystal structure significantly. They construct n-type (extra electrons) and

	p-type (extra holes) band diagrams and relate doping to the precise engineering that makes transistors and solar cells possible. The DQB is updated: learners can now partially explain the solar panel phenomenon. Lesson 5 (PhET Simulation — Semiconductor Lab): Learners use the PhET "Semiconductor" simulation to systematically vary doping type, doping concentration, and temperature, collecting I–V data and constructing evidence-based explanations for p-n junction behaviour. Forward and reverse bias are explored. Learners return to the anchoring phenomenon: why does high temperature reduce solar panel efficiency? Lesson 6 (Superconductors Explained): A dramatic supporting phenomenon (floating magnet video — Meissner effect) launches exploration of superconductivity. Learners investigate the concept of critical temperature, zero resistance, and the Meissner effect, contrasting superconductors with ordinary semiconductors using band theory and lattice vibration arguments. The DQB expands to a new question: could superconducting transmission lines eliminate Kenya's electricity distribution losses? Lesson 7 (Case Studies — Solar Panels, Microchips, Superconducting Magnets): Learners apply their cumulative understanding to three structured Kenyan-context case studies: (1) photovoltaic solar panels powering rural schools in Turkana County, (2) microchips inside M-PESA mobile payment handsets enabling financial inclusion, and (3) superconducting MRI magnets at Kenyatta National Hospital. Groups present findings, the DQB is finalised, and learners produce a revised annotated model explaining the original anchoring phenomenon in full. Career connections are made explicit.
Supporting Phenomena	– Germanium transistor radio: an old Sanyo transistor radio from the 1970s (common in Kenyan markets) stops working reliably when left in a hot matatu — heat destabilises the germanium transistors, hinting that temperature profoundly affects semiconductor behaviour in ways a simple resistor cannot explain. – Maglev demonstration — "The Floating Magnet": a small permanent magnet visibly levitates above a dish of liquid-nitrogen-cooled YBCO superconductor tile (video demonstration), defying learners' expectation that all materials always respond to gravity and magnetic fields the same way regardless of temperature. – LED vs. incandescent torch: a modern LED torch (semiconductor-based) remains cool and runs for days on one battery, while an old incandescent bulb torch (conductor-based filament) grows hot and drains the battery quickly — prompting questions about how electrons lose or conserve energy differently in different materials. – Kenya Power smart meter: the digital display on a modern prepaid electricity meter in a Nairobi household contains millions of transistors on a microchip the size of a fingernail — how can so much computational power fit in such a tiny space, and what makes the material special?

LESSON 1 (80 minutes (double lesson)): The Solar Panel Puzzle — Encountering the Phenomenon & Sorting Materials	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and establish productive confusion about why silicon — the material inside a solar panel — behaves differently from both conductors and insulators, and why heat reduces rather than increases its efficiency.
Knowledge	Learners will be able to: (a) classify common Kenyan materials as conductors, insulators, or semiconductors based on circuit-testing evidence; (b) describe the anchoring phenomenon — the Kajiado solar panel puzzle — in their own words; (c) state that semiconductors have electrical conductivity between that of conductors and insulators.
Skills	Learners will be able to: (a) design and conduct a simple conductivity-testing circuit to gather evidence about material properties; (b) record and interpret observations using a prediction–evidence–reasoning (PER) table; (c) construct an initial class Driving Question Board (DQB) by generating testable questions arising from the phenomenon.
Attitudes	Learners will appreciate that unexpected experimental results — such as silicon's in-between behaviour — are scientifically valuable and drive deeper inquiry, and will demonstrate intellectual honesty when their predictions are contradicted by evidence.
Key Inquiry Question	How do the electrical properties of materials depend on their atomic and energy band structure?
Purpose in Storyline	This is the opening lesson of the sub-strand. It anchors the entire unit in a real Kenyan context (a Kajiado farmer's solar panel), surfaces learners' prior models of conductors and insulators, and creates the productive confusion that motivates energy band theory. The Driving Question Board opened here will be revisited and expanded in every subsequent lesson.
Safety Notes	Ensure the conductivity-testing circuit uses a 1.5 V AA battery only — NOT a mains supply. Remind learners never to touch both terminals of a battery simultaneously with wet hands. Graphite pencil leads are fragile; handle with care to avoid eye injury from snapping. Dispose of any broken glass or sharp wire offcuts in the designated sharps container.

B. LESSON OVERVIEW
Learners are confronted with the anchoring phenomenon — a Kajiado farmer's silicon solar panel produces less power in hot afternoon sun than on a cool morning — through a short teacher-narrated story supported by a data table. This violates their intuition that 'more sunlight = more electricity'. Learners then predict whether familiar Kenyan materials (copper wire, plastic sufuria handle, graphite pencil, silicon chip/coin) are conductors or insulators, test each one using a simple 1.5 V circuit, and discover that silicon and graphite behave unexpectedly. The lesson closes by naming the new category — semiconductor — and opening the class Driving Question Board with learner-generated questions. No explanatory model is given yet; the lesson is intentionally designed to maximise curiosity and epistemic need for energy band theory in Lesson 2.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [The periodic table]	Learners are first shown a projected image of a silicon solar panel mounted on a simple	1. Display the Kajiado solar panel image and say: 'This is Mama Akinyi's solar panel in Kajiado. On	Think-Pair-Share with a Prediction Tally Wall — pairs commit to written predictions before any	Observable indicator: Every learner has completed the 'I predict… because…' sentence and

[- classification of elements](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Solar Power (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	irrigation frame in Kajiado County. The teacher narrates a short story (≈3 min) about Mama Akinyi, a farmer in Kajiado who notices her water pump runs strongly on cool mornings but weakly on hot afternoons, even though the sun is brighter. A two-row data table is displayed (Morning: 18 °C, partly cloudy, pump output 420 W; Afternoon: 35 °C, full sun, pump output 310 W). Learners write a one-sentence prediction in their notebooks: 'I predict this happens because…' Then, working in pairs, learners receive a Prediction–Evidence–Reasoning (PER) table listing six materials — copper wire, plastic mkono wa sufuria handle, sharpened graphite pencil lead, steel nail, rubber eraser, and a small silicon component (e.g., a diode or coin of silicon-rich material labelled 'silicon chip'). For each material, learners tick either 'Conductor' or 'Insulator' and write a one-sentence reason based on prior knowledge. Pairs share predictions with a neighbouring pair (Think-Pair-Share) before the teacher cold-calls 4 pairs to report predicted classifications to the class.	Monday morning at 18 °C her water pump produced 420 watts of power. On Monday afternoon at 35 °C — brighter sunshine, hotter day — her pump produced only 310 watts. Take 10 seconds of silence to sit with that.' [WAIT 10 seconds — no talking, let the cognitive dissonance settle.] 2. Say: 'In your notebook, complete this sentence in 15 seconds: I predict this happens because…' [WAIT 15 seconds.] 3. Distribute PER tables. Say: 'With your partner, predict whether each material is a conductor or an insulator. You have 4 minutes. Use what you already know — there are no wrong predictions yet.' 4. After 4 minutes, cold-call 4 pairs (select two high-confidence pairs and two hesitant pairs for range): 'Pair 3 — what did you predict for the graphite pencil lead and why?' Record predictions visibly on the board in a class tally WITHOUT correcting any errors. 5. Highlight any disagreements on the board and say: 'We have a disagreement about graphite and silicon. Good. That means we need evidence. Let's test.'	evidence is gathered, making prior conceptions visible and creating personal stakes in the outcome. Recording all predictions publicly (including incorrect ones) normalises scientific revision and sets up the 'surprise moment' when silicon is tested.	ticked conductor/insulator for all six materials on the PER table before testing begins. Teacher scans tables during cold-call phase. Red flag: any learner who has not written a reason — teacher prompts: 'What have you seen this material used for that gives you a clue?'
Observe Phase 📹 **VIDEO:** [The periodic table - classification of elements](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Solar Power (Flexbook)](#)	Groups of 3 receive a pre-assembled conductivity tester: a 1.5 V AA battery in a holder, two crocodile-clip leads, and a 3 V LED indicator (or small bulb). One lead is permanently attached to the positive terminal; learners touch the free ends of both leads to each material in turn and observe whether the LED lights up (conductor), does not light (insulator), or glows dimly (partial conductor). Each group tests all six materials and records: LED	1. Demonstrate the circuit once using the copper wire: 'Watch — I touch both clips to the copper wire. Full brightness. Copper is a conductor. Your job is to test the rest and record exactly what you see — not what you expect.' 2. Circulate during testing (≈12 min). Pause at any group where the silicon test is happening: 'What do you see? Is it fully on or fully off?' [WAIT 10 seconds for the group to articulate before moving on.]	Evidence Recording with Anomaly Spotlighting — learners record raw observations (LED brightness) rather than conclusions, then the teacher spotlights the anomaly (silicon's dim glow) to the whole class. This strategy focuses attention on the data that will drive the need for a new explanatory model (energy band theory), directly connecting the observation to the anchoring phenomenon (the solar panel is made of silicon).	Observable indicator: All groups have a completed Evidence column distinguishing LED-on, LED-dim, and LED-off results. Teacher checks that no group has recorded silicon as simply 'conductor' or 'insulator' without noting the dim/partial response. Red flag: groups who record 'conductor' for silicon because the LED lit at all — teacher redirects: 'Compare the brightness of the silicon test to the copper test. Are they the same? Record what you

Source: CK-12 🔍 Search ARES for similar readings	fully on / LED dim / LED off in the Evidence column of their PER table. Critically, the silicon component and sharpened graphite pencil lead both cause the LED to glow DIMLY — not fully on, not fully off. This is the key observation. Groups photograph or sketch the LED brightness for each material. A class results table is projected live as the teacher collates group findings by cold-calling each group.	3. After testing, project an empty class results table. Cold-call one reporter per group (rotate so different learners report): 'Group 2 — what did you observe for the graphite pencil lead? Describe the LED, not your conclusion.' 4. When the class table shows that silicon and graphite both gave DIM results, say: 'Interesting. Copper — full brightness. Rubber — off. Silicon — dim. The LED is speaking to us. What is it saying?' [WAIT 15 seconds — cold-call 3 individual learners.] 5. Say: 'This dim result does not fit our two boxes — conductor or insulator. We need a third category. Hold that thought.'		actually see.'
Explain Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners complete the Reasoning column of their PER table: 'My evidence matches / does not match my prediction because…' They then participate in a whole-class discussion. The teacher introduces the term SEMICONDUCTOR for the first time and writes a working definition on the board: 'A semiconductor is a material whose ability to conduct electricity is between that of a conductor and an insulator, and can change depending on conditions.' Learners are then asked to connect this to the Kajiado phenomenon: if silicon is a semiconductor and conditions affect its conductivity, which condition — temperature, light, or both — might explain why Mama Akinyi's panel performs differently in the morning versus the afternoon? Learners write a revised one-sentence explanation in their notebooks, then share with the class. The teacher does NOT	1. Give learners 3 minutes to complete the Reasoning column independently. [WAIT — do not talk during this time.] 2. Write on the board: SEMICONDUCTOR. Say: 'This is the name physicists give to materials like silicon and graphite that sit in between. Write this definition: A semiconductor is a material whose conductivity is between that of a conductor and an insulator and can change depending on conditions.' 3. Point to the projected Kajiado data table and say: 'Mama Akinyi's solar panel is made of silicon — a semiconductor. The panel works differently at 18 °C versus 35 °C. Using only what we have just observed — not what you've read — why might temperature matter for a semiconductor?' [WAIT 15 seconds.] Cold-call 4 learners by name for their revised explanation. 4. Record all four responses on the board verbatim under the	Claim-Evidence-Reasoning (CER) Writing — learners explicitly connect their observation (evidence) to the phenomenon (claim about the solar panel) through a reasoning step. By writing revised explanations and posting them as 'current best thinking' rather than right/wrong answers, the strategy builds epistemic humility and models authentic scientific practice, directly motivating the energy band model to come in Lesson 2.	Observable indicator: Each learner's revised explanation in their notebook explicitly mentions silicon, temperature or light, and the word 'semiconductor' or 'conductivity'. Teacher reads 6–8 notebooks during the DQB phase. Red flag: learners who write 'the panel gets too hot and melts' — teacher probes: 'What did the LED test show us about how silicon's conductivity changes? Can you connect that to the solar panel?'

	confirm or reject any explanation — instead, responses are added to the DQB as 'our current best thinking'.	heading 'Our Current Best Thinking'. Do not evaluate them. Say: 'All four of these could be right, partly right, or wrong. We will find out. But notice — none of our old ideas about simple conductors and insulators explains this. We need a new model.' 5. Say: 'To explain what we saw today and to answer Mama Akinyi's puzzle, we are going to need to look inside the silicon atom itself — at the energy levels of electrons. That is where we go next lesson.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or small pieces of paper). On the FIRST note they write one question that today's observations made them wonder about — something they cannot yet answer. On the SECOND note they write one thing they are now sure of. Notes are posted on the class DQB — a large sheet of paper or section of the board divided into two columns: 'Questions we are investigating' and 'Things we are confident about'. The teacher reads selected questions aloud, groups similar questions together (clustering), and labels emerging theme clusters (e.g., 'Why does temperature matter?', 'What is special about silicon's structure?', 'How does the solar panel actually work?'). The class agrees on the overarching Driving Question, which is written in large letters at the top of the board and remains visible throughout the unit.	1. Distribute two sticky notes per learner. Say: 'Two minutes — no talking. First note: one question today gave you that you cannot yet answer. Second note: one thing you are now confident about. Go.' [WAIT 2 minutes strictly.] 2. Invite learners row by row to post their notes. While they post, teacher reads questions aloud one at a time without commentary. 3. Cluster similar questions visibly in front of the class: 'These three questions are all asking about temperature and silicon — I'll group them here. These two are asking about what is inside silicon at the atomic level — they go here.' 4. Point to the overarching driving question already written at the top of the DQB: 'Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun than on a cool morning — and what is so special about silicon that it can act as both a conductor and an insulator depending on conditions?' Say: 'Every lesson from now until the end of this sub-strand, we will	Driving Question Board (DQB) — a living public record of the class's growing understanding that makes collective knowledge-building visible. By clustering learner-generated questions, the teacher helps learners see that their curiosity is coherent and scientifically tractable, not random. Returning to the DQB in every lesson creates continuity across the storyline and gives learners agency over the direction of inquiry.	Observable indicator: Every learner has posted at least one genuine question (not a statement) and one confidence claim. Teacher checks that questions are anchored to the phenomenon — not generic (e.g., 'Why does silicon conduct a little?' is anchored; 'What is electricity?' is too broad). Red flag: any learner who posts only statements or writes 'I don't know' — teacher sits with them briefly: 'What surprised you most in the circuit test today? Turn that surprise into a question.'

		return to this board. Some of your questions will get answered. You will add new ones. By the final lesson, we should be able to answer the big question at the top.' 5. Photograph the DQB for the class record and for the ARES platform.		
Model Building Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Solar Power (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	As a closing individual task, learners draw a simple Initial Model in their notebooks. They sketch three boxes side by side labelled CONDUCTOR (copper wire), INSULATOR (rubber eraser), and SEMICONDUCTOR (silicon chip). Inside each box they draw what they currently imagine the inside of each material looks like — using any symbols or pictures they choose — that would explain the LED brightness results they observed. There is no right answer at this stage; the model is a snapshot of each learner's current thinking. Below the boxes, learners write: 'My model cannot yet explain why heating silicon makes it conduct less efficiently. I need to learn about…' and complete the sentence. These Initial Model notebooks are kept and will be returned in Lesson 6 for learners to compare with their final models.	1. Say: 'Last task — individual, silent, 5 minutes. Draw your initial model. No textbooks, no phones. This is your thinking right now — it will be wrong in some ways and that is completely fine. Scientists start with imperfect models and improve them.' [WAIT — circulate silently, do not correct drawings.] 2. After 5 minutes, cold-call 2 learners to briefly describe (not show) their model to the class: 'Wanjiku, in one sentence, what did you draw inside the conductor box?' Acknowledge both without evaluating. 3. Say: 'Hold onto these. In Lesson 6, you will get them back and compare what you thought today with what you know by then. The difference is your learning.' 4. Close by returning to the phenomenon: 'We still cannot fully explain Mama Akinyi's puzzle. Tomorrow we go inside the silicon atom. Come ready to think about electrons and energy.'	Initial Model Documentation — making learners' pre-instructional mental models explicit and permanent creates a personal baseline that can be revisited and revised, making conceptual change visible and meaningful. The 'I need to learn about…' sentence starter explicitly links the model's inadequacy to the epistemic need that Lesson 2 will address, maintaining coherent storyline continuity.	Observable indicator: Each learner's Initial Model has three labelled boxes with at least one drawn feature inside each (dots, lines, shading — any representation counts) and a completed 'I need to learn about…' sentence that references either silicon's structure, electrons, temperature, or energy. Teacher collects and photographs a sample of 6 models to inform the depth of energy band theory introduction in Lesson 2. Red flag: any learner with empty boxes — teacher prompts: 'What does your circuit test tell you is different about these three materials? Draw that difference, even as a question mark.'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes: (1) Which materials produced the most productive confusion — did silicon and graphite both generate surprise, or did learners accept the dim LED result without curiosity? If learners were not surprised, the Phase 3 discussion may need more time in the next lesson to surface why this matters for the solar panel. (2) Review the DQB: are the learner questions genuinely anchored to the Kajiado phenomenon, or are they too abstract (e.g., 'What is electricity?')? If too abstract, begin Lesson 2 by re-reading the Kajiado data table and asking: 'Which of our DQB questions would help Mama Akinyi specifically?' (3) Scan the Initial Models collected: do learners draw any internal structure for the conductor versus insulator, or are all three boxes identical? If identical, Lesson 2's band diagram introduction should begin with a direct contrast — 'If they were the same inside, why did the LED behave differently for each?' (4)

Note any learner who expressed frustration at not getting an explanation today. Validate this in the Lesson 2 opening: 'You were right to be frustrated — today we finally get the model that explains it.' (5) Were all groups able to identify the dim LED result for silicon clearly, or did LED quality/battery strength vary? If results were inconsistent, consider using a multimeter to measure resistance of each material as a supplementary activity in Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Using a 1.5 V conductivity-testing circuit, we tested copper wire, plastic sufuria handle, graphite pencil lead, steel nail, rubber eraser, and a silicon chip. The LED was fully bright for copper and steel (conductors), fully off for plastic and rubber (insulators), and glowed DIMLY for graphite and silicon. This means silicon and graphite are neither full conductors nor full insulators.
What did I learn?	Materials that conduct electricity partially — with conductivity between conductors and insulators — are called SEMICONDUCTORS. Silicon is a semiconductor. Its conductivity can change depending on conditions such as temperature and light. This is why the solar panel on Mama Akinyi's farm in Kajiado behaves differently in the morning (18 °C) versus the afternoon (35 °C).
How does this explain the phenomenon?	We cannot yet fully explain WHY silicon's conductivity changes with temperature, or why this makes the solar panel less efficient when it is hotter. To explain this, we need a new model of how electrons are arranged inside solid materials — the energy band model. This will be our focus in Lesson 2.

LESSON 2 (40 minutes): Energy Bands and the In-Between World of Silicon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To build and use an energy band model that explains why silicon is neither a full conductor nor a full insulator, and to connect this model to the Kajiado solar panel phenomenon.
Knowledge	Learners will be able to describe the valence band, conduction band, and band gap for a conductor, an insulator, and a semiconductor; explain how band gap size determines electrical conductivity; and state why silicon occupies an in-between position on the conductivity spectrum.
Skills	Learners will construct annotated band diagrams for three classes of materials, interpret given diagrams, and use evidence from the previous lesson sorting activity to justify the placement of silicon in the semiconductor category.
Attitudes	Learners will appreciate that models in physics are built to explain real observations, and that productive confusion — such as silicon behaving unexpectedly — is the starting point of genuine scientific understanding.
Key Inquiry Question	How do the electrical properties of materials depend on their atomic and energy band structure?
Purpose in Storyline	This lesson provides the theoretical model learners need to begin explaining the Kajiado solar panel puzzle. Lesson 1 created the puzzle by showing that silicon is neither conductor nor insulator. Lesson 2 resolves the first layer of confusion by introducing energy band theory, giving learners a new explanatory tool they will extend through Lessons 3 to 7.
Safety Notes	No chemical or electrical hazards in this lesson. If physical band diagram cards are used, ensure scissors are handled safely. Learners with visual impairments should be given tactile or high-contrast versions of the band diagrams.

B. LESSON OVERVIEW

Learners begin by revisiting the sorting results from Lesson 1 — particularly the unexpected in-between behaviour of silicon and graphite — and acknowledge that Ohm's Law alone cannot explain what they saw. The teacher introduces energy band theory using a Kenyan market analogy (shoppers on stairs versus an open field), and learners construct annotated band diagrams for a metal (copper wire), an insulator (plastic handle), and a semiconductor (silicon chip). They compare band gap sizes and link these directly to conductivity. The lesson closes by returning to the Kajiado solar panel: learners use their new model to make a first partial explanation of why silicon can conduct electricity at all, updating the Driving Question Board with refined questions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English	Learners open their notebooks and view the Lesson 1 summary table on the board. The teacher re-displays the circuit test results: copper conducted easily, plastic did not conduct, graphite gave a	Display the prompt on the board and read it aloud. Say: 'You do not need the correct answer yet. I want to see what model you are already building in your mind. Every idea is worth writing down.'	Elicitation of prior models — making learner thinking visible before instruction so misconceptions can be addressed explicitly. Recording ideas publicly signals that scientific thinking	Teacher scans written responses during circulation, noting whether learners think in terms of: (a) electron quantity, (b) electron freedom of movement, or (c) energy needed. This diagnostic

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	dim glow, and silicon gave a faint flicker only in bright light. Each learner independently writes a response to this prompt: 'In Lesson 1 we said electrons need to move to create current. For the copper wire, something made it easy for electrons to move. For the plastic handle, something stopped them completely. For silicon, what do you think is different about how electrons are held inside the material? Draw or write your idea — any model is valid at this stage.' Learners have 3 minutes to write without discussion, then share with a partner for 1 minute.	Set a visible 3-minute timer. Circulate silently — do not correct or affirm yet. After pair share, cold-call two or three learners: 'Amani, tell me one idea your partner had that surprised you.' Record three to four different learner ideas on the board in a column labelled LEARNER MODELS — these will be revisited at the end of the lesson. WAIT TIME: after each cold-call response, wait at least 5 seconds before speaking, allowing other learners to process.	starts with honest uncertainty.	informs how much scaffolding to provide during the Explain phase.
Observe Phase 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects or draws a simple staircase diagram — labelled as a Nairobi market with two floors. Ground floor is labelled VALENCE BAND (electrons at rest, bound to atoms). Upper floor is labelled CONDUCTION BAND (electrons free to move and carry current). The gap between floors is labelled BAND GAP. The teacher then shows three versions of this diagram side by side: (1) Metal — the two floors overlap so electrons flow freely between them at all times. (2) Insulator — the gap between floors is very tall (like a three-storey climb with no stairs) so virtually no electron can make the jump at room temperature. (3) Semiconductor — the gap is small (a single step) so a few electrons can jump across with a small push of energy such as heat or light. Learners observe quietly for 90 seconds, then answer in their notebooks: 'Which diagram matches copper? Which matches plastic? Which matches silicon? How do you know?'	Draw or project each diagram slowly, narrating as you go: 'Imagine electrons are shoppers on the ground floor of Gikomba Market. To carry current — to do useful work — they need to reach the upper floor where they can move freely. For copper, there is no wall between floors — shoppers move up and down without effort. For plastic, the upper floor is three storeys above with no staircase — almost no shopper can reach it. For silicon, there is a small step — most shoppers stay below, but if a shopper gets a push — from sunlight, from heat — some can jump up.' Point explicitly to the band gap on each diagram. Say: 'The size of this gap is the single most important idea in this lesson.' WAIT TIME: After drawing all three, pause for 30 seconds of silent observation before asking learners to write.	Concrete analogy bridging (market staircase to band diagram) followed by immediate application — learners must map the analogy back onto familiar materials from Lesson 1. This reinforces that the model is built to explain real observations.	Teacher collects or scans notebook responses. Correct matching (copper = overlap, plastic = large gap, silicon = small gap) with a reason demonstrates readiness for model construction. Learners who match correctly but cannot state a reason need targeted support during Model Building.

Explain Phase 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Working in groups of three, learners receive a blank band diagram template (three side-by-side boxes with a horizontal dividing space). Each group labels and annotates all three diagrams — conductor, insulator, semiconductor — using the following required labels: valence band, conduction band, band gap, arrow showing electron jump for semiconductor, and a note on whether the band gap is large, zero/overlapping, or small. Groups then answer two focus questions in writing: (1) 'A material has a band gap of 1.1 electron-volts — roughly the energy of visible light. What class of material is this? How does this fact connect to a solar panel?' (2) 'Why can plastic never conduct electricity no matter how much sunlight hits it?' Each group nominates a spokesperson to share one answer with the class.	Distribute the blank templates. Say: 'Your job as a group is to build the model — not just copy it. Every label must be in your own words on the diagram.' Circulate and use targeted questions: 'What does the arrow on the semiconductor diagram represent physically — what is actually moving?' and 'If I made the band gap even smaller, what would happen to conductivity?' For the focus questions, prompt deeper thinking: 'Question one connects directly to why Mama Akinyi in Kajiado has a silicon solar panel and not a copper one or a plastic one. Think about that.' After spokesperson sharing, address the whole class: 'Notice that 1.1 eV matches the energy of visible light photons — this is not an accident. Silicon was chosen for solar panels precisely because its band gap matches sunlight energy. This is engineering using physics.' WAIT TIME: After each spokesperson, wait 6 seconds before responding, allowing peer evaluation.	Collaborative model construction followed by application questions that bridge back to the anchoring phenomenon. The focus questions are deliberately sequenced — question one anchors forward to solar cells, question two anchors backward to the insulator evidence from Lesson 1.	Circulate and use a mental checklist: Does the diagram show all five required labels? Is the semiconductor band gap drawn noticeably smaller than the insulator gap? Does the electron-jump arrow appear only on the semiconductor diagram? Groups missing two or more elements receive a targeted prompt card with a corrective question rather than a direct answer.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook)	The teacher projects the class Driving Question Board started in Lesson 1. Learners individually revisit the Kajiado solar panel phenomenon using the new model. Each learner writes on a sticky note — or in their notebook if physical notes are unavailable — a response to this sentence stem: 'Now that I know about energy bands, I can partially explain... but I still wonder...' The first part should reference the band model. Learners add their notes to the DQB under two columns: EVIDENCE WE NOW HAVE and QUESTIONS WE STILL NEED TO	Read the driving question aloud: 'Why does a silicon solar panel on a Kajiado farm produce less power in hot afternoon sun than on a cool morning — and what is special about silicon?' Say: 'Two lessons ago this question had no answer. Today you have built a model. What part of this question can your model now touch?' After learners write, cluster the sticky notes publicly, narrating the groupings: 'Several of you can now explain why silicon conducts at all — that belongs in the EVIDENCE column. Several of you are asking why heat makes it worse, not better —	Public DQB update consolidates learning and makes the boundary of current knowledge visible. Clustering questions helps learners see that their confusion is shared and that the storyline is designed to address each cluster in turn.	Quality of the EVIDENCE entries reveals whether learners can connect band theory to the phenomenon. Entries that say only 'silicon has a small band gap' without linking to electron movement or photon energy indicate surface-level engagement and flag learners needing reinforcement in Lesson 3.

Source: CK-12 🔍 Search ARES for similar readings	ANSWER. The teacher reads selected entries aloud, clustering similar questions together.	that question goes in the STILL WONDERING column and we will investigate it in Lesson 3.' Say explicitly: 'A good scientist does not pretend the model answers everything. Knowing the boundary of your model is a skill.' WAIT TIME: After reading each cluster aloud, pause 4 seconds before moving to the next.		
Model Building Phase 📹 **VIDEO:** Global winds and currents Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner individually draws a final annotated model in their science notebook — a single diagram of the silicon band structure at room temperature with the following annotations: (1) valence band with electrons shown as filled circles, (2) conduction band mostly empty, (3) band gap labelled 1.1 eV with a note: this matches visible light energy, (4) a dotted arrow showing one electron jumping across when light or heat energy is absorbed, (5) a caption connecting the diagram to the Kajiado farm: 'This is why Mama Akinyi sees electricity from the panel — sunlight gives enough energy to push electrons across the silicon band gap. But we still do not know why more heat means LESS efficiency.' Learners swap notebooks with a partner and use a peer-review checklist (five items matching the five annotations) to verify completeness.	Say: 'This diagram is your personal evidence artefact for Lesson 2. It must be your own work. Your annotation in item 5 is the most important — it shows you know the model is not finished yet.' Circulate during drawing. For learners who draw the electron jump going downward (a common error), ask: 'If the electron is going from the valence band to the conduction band, which direction on the diagram is that — up or down? Why?' After peer review, ask three learners at random to read their caption for item 5 aloud. Say: 'Notice that every caption ends with a question. That question is what drives Lesson 3. Hold onto it.' Collect or photograph two or three notebooks as formative evidence for teacher reflection. WAIT TIME: After the third learner reads their caption, hold 8 seconds of silence before closing the lesson — allow the unresolved question to sit.	Individual model construction followed by structured peer review ensures every learner produces a usable artefact. The deliberate open-ended caption embeds forward momentum into the lesson close, sustaining curiosity rather than providing false closure.	Peer-review checklists are collected by the teacher. Any item checked as incomplete flags a specific gap. The caption in item 5 is the highest-value formative data point: it reveals whether the learner understands the model is partial and can articulate the remaining question in their own words.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did learners draw the semiconductor band gap noticeably smaller than the insulator gap, or did they draw all gaps the same size? If the latter, the analogy needs a more physical anchor in Lesson 3 — consider using actual ruler measurements on the board. (2) Did learners connect the 1.1 eV band gap value to silicon being chosen for solar panels, or did they treat it as an arbitrary fact? If the connection was weak, open Lesson 3 with a direct callback. (3) Were the DQB STILL WONDERING entries focused on temperature effects? If most questions were about something else entirely, the lesson narrative may not have foregrounded the thermal puzzle enough — adjust the Lesson 3 hook accordingly. (4) Which learners did not write a complete caption for item 5? Plan a brief paired

catch-up at the start of Lesson 3 for those learners before whole-class instruction begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In our circuit tests from Lesson 1, silicon allowed only a faint flicker of current — less than copper but more than plastic. Today we looked at the band diagram for silicon and saw that its band gap is small but not zero — about 1.1 eV. A few electrons can jump across when they receive energy from light or heat.
What did I learn?	All solid materials have a valence band (where electrons are normally held) and a conduction band (where electrons can move freely). The size of the band gap determines conductivity. Metals have overlapping bands — no gap. Insulators have a very large gap. Semiconductors like silicon have a small gap, matched to the energy of visible light photons, which is why a solar panel made of silicon can convert sunlight into electricity.
How does this explain the phenomenon?	We can now explain why silicon conducts electricity at all: sunlight provides enough energy to push electrons across the small silicon band gap into the conduction band, where they can flow as electric current. This is the first layer of the Kajiado solar panel explanation. We cannot yet explain why high afternoon temperatures reduce power output — that puzzle will be investigated in Lesson 3 using the idea of electron-hole pairs in pure silicon.

LESSON 3 (70 minutes): Intrinsic Semiconductors — The Energy Band Model and Silicon's Dual Identity	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explain why silicon behaves as neither a pure conductor nor a pure insulator by applying the energy band model, and to use this model as the first complete explanation for why the Kajiado solar panel produces less power on a hot afternoon than on a cool morning.
Knowledge	Learners explain the concept of energy bands (valence band, conduction band, band gap) in solids; distinguish conductors, insulators, and intrinsic semiconductors using band diagrams; describe how thermal energy promotes electrons across the band gap in silicon, generating electron-hole pairs; and state how increasing temperature increases charge carrier density but simultaneously reduces carrier mobility, leading to net power loss in a silicon solar panel.
Skills	Learners construct and annotate energy band diagrams for a conductor, an insulator, and an intrinsic semiconductor; interpret data tables comparing band-gap energies of different materials; analyse a temperature-vs-power graph for the Kajiado solar panel and relate it to the band model; and evaluate evidence to revise a scientific model.
Attitudes	Learners appreciate that a deeper model of matter (energy bands) is necessary when everyday observations — like the solar panel working better in the cool morning — cannot be explained by previous knowledge of simple circuits; and they value precision and evidence-based reasoning in constructing scientific explanations.
Key Inquiry Question	How do the electrical properties of materials depend on their atomic and energy band structure?
Purpose in Storyline	In Lessons 1–2, learners established that silicon is a semiconductor with conductivity between conductors and insulators, and they observed the puzzling temperature-power trend in the Kajiado solar panel data. In this lesson — the conceptual pivot of the sequence — learners acquire the energy band model as the mechanistic tool that explains silicon's dual identity. This directly sets up Lesson 4, where doping (n-type and p-type) will be layered onto the band model to explain how engineers harness and refine silicon's properties for transistors and solar cells.
Safety Notes	No hazardous chemicals or high-voltage equipment are used in this lesson. If printed band-diagram cards are cut with scissors, learners should exercise standard care. Ensure the classroom projector or screen is positioned so all learners can read energy-level labels clearly without straining.

B. LESSON OVERVIEW
Learners begin by revisiting their Lesson 1–2 predictions and observations about why the Kajiado solar panel underperforms on hot afternoons. They are challenged to move beyond the surface observation ('less power when hot') toward a mechanistic explanation. Through a guided data-analysis activity using a band-gap reference table and a temperature-vs-power graph for silicon solar panels, learners are introduced to the energy band model. Using physical card-sorting and diagram-construction activities, they build and annotate band diagrams for a conductor (copper wire used in Nairobi power lines), an insulator (rubber used to coat those wires), and an intrinsic semiconductor (silicon). They then apply the model to explain the phenomenon: at higher temperatures, more electron-hole pairs are generated (increasing carrier density) but increased lattice vibrations scatter carriers more strongly (reducing mobility), and the net effect is a drop in open-circuit voltage and panel efficiency. The lesson closes with DQB updating and a cumulative model revision, leaving learners poised to ask: 'If doping can be used to control carrier concentration deliberately, can engineers overcome the temperature problem?'

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two pieces of data side by side on the board: (A) a photograph of the Kajiado farm solar panel under a blazing afternoon sun at 35 °C, and (B) the same panel on a cool overcast morning at 18 °C. Alongside each image is a single data point — 'Afternoon power output: 210 W' and 'Morning power output: 310 W'. Learners individually write a two-sentence prediction in their exercise books answering: 'Why do you think silicon produces less power when it is hotter? Use what you already know about electrons and materials.' After 3 minutes of silent writing, learners share predictions in pairs (Think-Pair-Share). Selected pairs share aloud. Predictions are recorded verbatim on the left column of the class Driving Question Board under the heading 'Our Best Guess — Lesson 3 Start'.	Display both images and data points using the projector. Read aloud: 'On Tuesday afternoon in Kajiado, this panel made 210 watts. On Wednesday morning — the very next day — it made 310 watts. Same panel. Same farm. Different time and temperature.' Pause for 12 seconds of WAIT TIME. Then say: 'In your exercise books, write your best prediction. Two sentences only. You have three minutes — start now.' Circulate silently, scanning books for three contrasting prediction types (e.g., 'electrons move faster when hot so more power', 'heat damages the panel', 'something to do with the material itself'). After sharing, cold-call exactly 3 pairs — choose one who wrote a 'more heat = more energy = more power' misconception, one who wrote about material properties, and one who expressed uncertainty. For each, say: 'Thank you — hold that idea. We will test it against evidence today.' Do NOT correct predictions yet.	Think-Pair-Share with verbatim DQB recording. This strategy makes existing mental models visible before instruction begins. By recording predictions verbatim, learners have a public stake in the explanation and a concrete reference point to revise by the end of the lesson. The deliberate selection of contrasting predictions (including the common misconception) creates productive cognitive dissonance that motivates the energy band model.	Observable indicator: Every learner has written at least two sentences in their exercise book before pair discussion begins. The teacher identifies the prevalence of the 'more heat = more power' misconception during circulation — if more than 60% of the class holds this view, the teacher notes to spend extra time on the mobility-vs-carrier-density trade-off in Phase 3.
Observe Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy	Activity A — Band-Gap Data Table Analysis (15 min): Each group of 4 receives a printed data card showing band-gap energies (in eV) for six materials: copper (0 eV — overlapping bands), aluminium (0 eV), silicon (1.12 eV), germanium (0.67 eV), diamond (5.47 eV), and rubber/polyethylene (~8 eV). Groups also receive a simple number line drawn on paper labelled 'Band-Gap Energy (eV)' from 0 to 9. They physically place labelled sticky notes for each	Distribute printed data cards and graph sheets before the lesson begins to save time. For Activity A, say: 'Your job is to be materials scientists at KIRDI — Kenya Industrial Research and Development Institute — in Nairobi. You are classifying materials for use in solar panel manufacturing. Place every material on your number line and decide: where does the semiconductor zone start and end? You have 8 minutes.'	Data-to-Model Bridging using a physical number line and a real performance graph. The number line makes abstract band-gap values concrete and spatially organised, allowing learners to 'see' the semiconductor zone as genuinely intermediate. The temperature-power graph grounds the lesson firmly in the anchoring phenomenon (Kajiado farm) with real numbers, preventing the energy band model from feeling purely theoretical. The video	Observable indicator: Each group's sticky-note number line correctly places silicon and germanium between copper/aluminium and diamond/rubber. The teacher checks that learners answer Q3 correctly (voltage drop dominates) — incorrect answers signal a need for re-teaching in Phase 3. Learner sketches of the band diagram from the video should show at least two horizontal bands with a gap between them; absence of the gap indicates the core concept has not

Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	material on the number line and circle the 'semiconductor zone' they think exists between conductors and insulators. Activity B — Temperature-Power Graph Reading (10 min): Groups are given a printed graph showing power output (W) on the y-axis versus panel temperature (°C, from 15 °C to 55 °C) on the x-axis for a typical silicon solar panel. Three trends are annotated on the graph: (1) open-circuit voltage decreases with temperature, (2) short-circuit current increases slightly with temperature, (3) net power output decreases overall. Learners answer three guiding questions written on the board: 'Q1 — At what temperature does power peak? Q2 — By roughly how many watts does power drop between 18 °C and 35 °C (as seen in Kajiado)? Q3 — Which trend dominates — the voltage drop or the current rise?' Learners record answers in their books. Activity C — Video Clip (5 min): Learners watch a short curated YouTube segment (referenced in the ARES platform resource library under 'Semiconductors and Applications') showing animated visualisation of electrons jumping from the valence band to the conduction band in silicon when heat is applied, and the concept of 'holes' moving in the opposite direction. Teacher pauses the video at the band diagram frame and asks learners to sketch what they see.	Circulate. After 8 minutes, cold-call 2 groups to share their number-line placements. For Activity B, project the graph on the board alongside the printed copies. Say: 'This is real published data for silicon panels — the same type used in Kajiado. Read the graph carefully and answer the three questions. You have 7 minutes.' After 7 minutes, cold-call 3 individual learners (one per question) and record their answers on the board. For Q3, allow 10 seconds of WAIT TIME before cold-calling. If learners cannot identify the dominant trend, say: 'Look at the steepness of the two curves — which changes faster as temperature rises?' For Activity C, before playing the video say: 'As you watch, focus on one thing: what happens to electrons in silicon when temperature increases? Sketch the band diagram you see on screen.' Pause the video at the 45-second mark showing the band diagram and give 2 minutes for sketching.	provides a dynamic visualisation that bridges the static diagram and the physical process.	yet been received.
Explain Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan	Activity A — Band Diagram Card Sort and Construction (15 min): Each group receives a set of 12 pre-cut cards bearing labels: 'Conduction Band (empty)', 'Conduction Band (partially filled)',	For Activity A, say: 'You are building the most important diagram in all of semiconductor physics. Every transistor in every phone, every solar cell on every rooftop in Kenya — all of it starts	Card Sort for Model Construction: Physical manipulation of labelled cards before drawing forces learners to make decisions about relationships between concepts (band position, gap size, material	Observable indicator: In copied band diagrams, every learner's silicon column shows (1) a valence band, (2) a clearly labelled band gap of ~1 eV, and (3) a conduction band with a small number of

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'Valence Band (full)', 'Valence Band (partially filled)', 'Band Gap: 0 eV', 'Band Gap: 1.12 eV', 'Band Gap: 5.47 eV', 'Free electrons at room temperature: many', 'Free electrons at room temperature: very few', 'Free electrons at room temperature: none without thermal energy', 'Material: Copper wire (Nairobi power line)', 'Material: Rubber insulation', 'Material: Silicon solar cell (Kajiado)'. Groups arrange cards into three columns (one per material) to build three complete band diagrams on their desk. They then copy the three diagrams into their exercise books with full labels and a one-sentence explanation below each. Activity B — Applying the Model to the Phenomenon (10 min): Teacher leads a whole-class structured discussion using the sequence: (1) 'In the cool morning at 18 °C, how many electrons in the silicon have enough thermal energy to jump the 1.12 eV band gap? A few or many?' (answer: a few — just enough for useful conductivity). (2) 'At 35 °C in the hot afternoon, does the number of electrons jumping the gap increase or decrease?' (answer: increases — more thermal energy available, more electron-hole pairs). (3) 'If more electrons are jumping, why does the panel produce LESS power — not more?' This is the conceptual crux. Teacher introduces the concept of carrier mobility: 'The silicon crystal lattice is vibrating more violently at 35 °C — like a crowded Nairobi bus during rush hour, electrons collide more and cannot move as freely. This reduces voltage more than the extra carriers can compensate.' Learners write a three-sentence	with this diagram. Take 12 minutes.' Circulate and specifically check that the silicon column shows a non-zero band gap AND partially filled conduction band (not empty). If a group draws the silicon conduction band as completely empty, crouch to their level and ask quietly: 'At room temperature, does silicon conduct a little or not at all?' — do not give the answer; let them reason. After 12 minutes, cold-call 1 group to 'present' their silicon diagram to the class while the teacher annotates a master diagram on the board. For Activity B, use the Socratic sequence as written. At question (3) — 'why does the panel produce LESS power?' — enforce 15 seconds of WAIT TIME before accepting any responses. Do not accept a hand until the full 15 seconds have elapsed. Say explicitly: 'I will wait the full 15 seconds. Everyone must think.' After responses, use the Nairobi bus analogy as written. Write the five-step causal chain on the board as learners dictate it sentence by sentence. For Activity C, say: 'One sentence only — you have 90 seconds.' Cold-call 2 learners to share. Accept all reasonable answers and say: 'We will come back to this question in Lesson 4 when we learn how engineers actually tune conductivity.'	identity) before committing pen to paper. This reduces cognitive overload and reveals misconceptions in the arrangement phase rather than in the written product. The Socratic sequence in Activity B uses guided questioning to build the causal chain incrementally, ensuring the counter-intuitive result (more heat = less power) is understood mechanistically, not merely memorised. The Nairobi bus analogy provides a culturally grounded mental image for the abstract concept of reduced carrier mobility.	electrons shown as dots. The five-step causal chain in learners' books is the primary evidence of conceptual understanding — the teacher reads 3–4 books during the DQB phase to check for accuracy. Learners who write only 'heat reduces power' without a mechanistic chain are flagged for targeted follow-up.

	causal chain in their books: 'Temperature increases → lattice vibrations increase → carrier mobility decreases → open-circuit voltage drops → net power output falls.' Activity C — Band Gap Engineering Teaser (3 min): Teacher briefly shows that germanium (0.67 eV) and gallium arsenide (1.42 eV) have different band gaps and asks: 'If you could choose the band-gap energy of a solar cell material, what value would you choose for a farm in Kajiado and why?' Learners write one sentence. This seeds Lesson 4's discussion of doping and engineering.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives one sticky note. They write a response to the prompt: 'Using the energy band model, write ONE sentence that partially answers our driving question: Why does the Kajiado solar panel produce less power on a hot afternoon?' Learners post their sticky notes on the class DQB under the column 'What We Can Now Explain — Lesson 3'. The class then reviews the DQB as a whole. The teacher reads three sticky notes aloud (one strong explanation, one partially correct, one that reveals a remaining gap). Together, the class identifies what the band model CAN now explain and what it CANNOT yet explain. Teacher records on the DQB: 'Still unexplained: How do engineers deliberately increase conductivity without just heating silicon? Why do solar cells have a built-in electric field that pushes electrons in one direction?' These become the motivating questions for Lesson 4.	Distribute sticky notes. Say: 'You have exactly 2 minutes to write one precise sentence using the words: band gap, carrier mobility, and voltage. Go.' After posting, read three notes aloud without naming authors. For the partially correct note, say: 'This is close — what is missing from this explanation?' Allow 10 seconds WAIT TIME, then cold-call 2 learners. For the note that reveals a gap (e.g., one that says 'silicon just gets tired in the heat'), say warmly: 'This is an honest record of where our thinking is right now — and that gap is exactly what drives our next investigation.' Write the two 'still unexplained' questions on the DQB in a different colour. Say: 'These two questions are your homework for your mind tonight. We will answer them in Lesson 4.'	Public DQB Posting with Gap Identification: Making individual understanding public and then collectively identifying explanatory gaps serves two functions — it consolidates lesson learning through articulation, and it generates genuine curiosity about the next lesson. The explicit identification of 'what we cannot yet explain' models authentic scientific practice, where answering one question always reveals deeper questions. This is directly analogous to how researchers at institutions like the University of Nairobi's Physics Department work.	Observable indicator: Sticky notes are assessed on a 3-point scale — (3) uses all three required vocabulary terms correctly in a causal sentence, (2) uses two terms or uses all three but without causal logic, (1) writes a descriptive statement without mechanistic vocabulary. The teacher records the distribution of scores to inform the opening of Lesson 4.

Model Building Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Introduction to Energy Resources (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative 'Silicon Solar Panel Model' booklet (a folded A4 sheet maintained across all 7 lessons). In Lesson 3's designated box, they draw and annotate an updated model that now includes: (1) a band diagram for silicon at 18 °C (cool morning in Kajiado) showing few electrons in the conduction band and a label 'higher mobility → higher voltage → more power'; (2) a band diagram for silicon at 35 °C (hot afternoon) showing more electrons in the conduction band but with a label 'lower mobility due to lattice vibrations → lower voltage → less power'; (3) an arrow connecting both diagrams to the real-world observation: 'Morning: 310 W output / Afternoon: 210 W output.' Beneath the model, learners write a 'Model Limitation' note: 'This model does not yet explain how the solar cell creates a one-way electric field that drives current out of the panel.' This limitation statement is a scaffold for Lesson 4's investigation of p-n junctions and doping.	Say: 'Open your model booklets to the Lesson 3 box. You have 7 minutes to draw and annotate your updated model. Your model must show both temperatures, both band diagrams, and connect them to the power numbers from Kajiado.' Circulate and look specifically for: learners who draw identical band diagrams for both temperatures (prompt: 'What is different about the number of electrons in the conduction band at each temperature?'); learners who show the higher temperature panel producing more power (prompt: 'Check your graph from Activity B — which temperature gives more power?'). With 2 minutes remaining, say: 'Now add the Model Limitation sentence — what does your model not yet explain?' Cold-call 1 learner to read their limitation aloud. Close by saying: 'In Lesson 4, we will add one more layer to this model — doping — and your limitation will become your first question.'	Cumulative Model Revision Booklet: Maintaining a running visual model across all 7 lessons externalises conceptual growth and makes the iterative nature of scientific modelling tangible. The explicit 'Model Limitation' requirement prevents premature closure — learners must articulate what their model cannot yet do, which is a core NGSS Science and Engineering Practice (Developing and Using Models, and Constructing Explanations). This practice mirrors how semiconductor engineers at companies like Solinc East Africa (a Kenyan solar manufacturer) iteratively refine their understanding of silicon behaviour.	Observable indicator: The completed model booklet entry shows two visually distinct band diagrams (different electron populations in the conduction band) correctly labelled with temperatures and power values, AND includes a model limitation statement that references the one-way current direction or the p-n junction concept. The absence of a limitation statement indicates a learner who believes the model is complete — a key misconception to address at the start of Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the majority of learners successfully articulate the counter-intuitive causal chain — that more thermal energy produces more carriers but reduces mobility and therefore net power? If not, the 'Nairobi bus' analogy for carrier mobility may need to be replaced with a physical demonstration (e.g., learners trying to walk quickly through a tightly packed corridor vs. an empty one). (2) Were the card-sort band diagrams completed accurately, or did learners persistently confuse the silicon and conductor diagrams? If confusion was widespread, consider pre-drawing the copper band diagram as a worked example before the card sort. (3) How strong were the DQB sticky-note explanations? If fewer than half the class scored a 3 on the formative rubric, plan a 10-minute re-teaching slot at the start of Lesson 4 before introducing doping. (4) Did the lesson run within 70 minutes? If Activities A and B in Phase 2 consumed too much time, the video clip (Activity C) can be assigned as an asynchronous ARES platform viewing task before Lesson 4. (5) Were there gender patterns in cold-call participation? Review your call log and ensure that female learners were represented equally in the three cold-call slots across all phases — this directly addresses the gender disparity PCI embedded in the KICD curriculum.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners analysed a temperature-vs-power graph for a silicon solar panel (based on the Kajiado farm phenomenon), observed that power output drops as temperature rises from 18 °C to 35 °C, and noted from a band-gap data table that silicon occupies an intermediate energy gap (~1.12 eV) between conductors (0 eV) and insulators (~8 eV).
What did I learn?	Learners learned that solids can be classified by their energy band structure: conductors have overlapping or partially filled bands with no gap; insulators have a large band gap that electrons cannot cross at room temperature; intrinsic semiconductors like silicon have a moderate band gap (~1.12 eV) that allows a small number of electrons to be thermally promoted to the conduction band, creating electron-hole pairs. Increasing temperature increases carrier density but decreases carrier mobility due to lattice vibrations, causing a net drop in open-circuit voltage and panel power output.
How does this explain the phenomenon?	Using the energy band model, learners can now partially explain the anchoring phenomenon: the Kajiado solar panel produces less power on a hot afternoon (35 °C) than on a cool morning (18 °C) because, although higher temperature promotes more electrons across silicon's band gap, it simultaneously increases lattice vibrations that scatter charge carriers, reducing their mobility and lowering the panel's output voltage more than the slight current increase can compensate. Silicon's band gap of 1.12 eV gives it its semiconductor identity — it is neither a free conductor like the copper in Nairobi's power lines nor an insulator like the rubber coating those lines.

LESSON 4 (80 minutes): Doping Silicon: How Impurities Turn a Crystal into a Switch	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explain how controlled introduction of impurities (doping) into pure silicon creates n-type and p-type semiconductors, enabling the directional flow of current that makes solar panels and other electronic devices possible.
Knowledge	Learners will be able to: (a) define doping and explain why it is used to modify semiconductor conductivity; (b) distinguish between n-type semiconductors (donor impurities, e.g. phosphorus) and p-type semiconductors (acceptor impurities, e.g. boron); (c) describe how free electrons and holes act as charge carriers in doped silicon; (d) explain how doping concentration affects conductivity.
Skills	Learners will be able to: (a) analyse diagrams of crystal lattice structures to identify donor and acceptor atoms; (b) construct and annotate visual models of n-type and p-type silicon; (c) interpret data tables comparing conductivity of pure and doped silicon at different temperatures; (d) use evidence to revise prior explanations of the Kajiado solar panel phenomenon.
Attitudes	Learners will: (a) appreciate how precise engineering at the atomic scale has transformed everyday Kenyan life (mobile phones, solar panels, M-PESA hardware); (b) demonstrate curiosity and persistence when working with abstract atomic models; (c) value accuracy and evidence-based reasoning when constructing scientific explanations.
Key Inquiry Question	How do the electrical properties of materials depend on their atomic and energy band structure?
Purpose in Storyline	In Lessons 1–3 learners established that silicon is a semiconductor whose conductivity increases with temperature due to electrons gaining enough energy to cross the band gap. However, pure (intrinsic) silicon is still a poor conductor at room temperature. Lesson 4 addresses the central engineering question: how do solar panel manufacturers make silicon conduct electricity reliably and in a controlled direction? By understanding doping, learners gain the conceptual tool needed for Lesson 5, where they will explore the p-n junction and finally explain why the Kajiado panel loses efficiency in the hot afternoon sun.
Safety Notes	No hazardous materials are used in this lesson. All activities are paper- and discussion-based. Remind learners to handle printed diagrams and scissors carefully if cutting activities are used. Ensure all learners have adequate seating and lighting for close diagram work.

B. LESSON OVERVIEW
Learners build on their band-gap model of intrinsic silicon (Lessons 1–3) to investigate how deliberate impurity atoms — a process called doping — dramatically alter silicon's conductivity. Using annotated crystal-lattice diagrams, a structured reading extract, a class data-analysis task, and a collaborative model-building activity, learners construct evidence-based explanations for n-type and p-type semiconductors. They connect doping directly to the Kajiado solar panel phenomenon by recognising that the silicon in a real panel is not pure — it is a precisely engineered p-n junction — and they begin to ask why the junction, rather than temperature alone, controls current direction. The lesson ends with learners updating the Driving Question Board and their cumulative explanatory model, preparing them for the PhET simulation in Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a 'Silicon Crystal Prediction Card' showing two diagrams side-by-side: (A) a pure silicon lattice with every atom sharing 4 covalent bonds and (B) the same lattice but with one phosphorus atom (5 valence electrons) substituted in the centre. Individually (3 min silent think), learners write answers to two questions printed on the card: (1) 'What happens to the extra electron that phosphorus brings — where does it go?' and (2) 'Will this silicon-with-phosphorus conduct electricity better or worse than pure silicon? Explain using your band-gap model from Lesson 3.' Learners then share predictions with their shoulder partner (2 min), before three pairs are cold-called to share with the class.	Distribute Prediction Cards face-down. Say: 'Before you flip the card, I want you to remember the band-gap model we built last lesson. We said electrons in pure silicon are trapped in the valence band unless they get enough energy to jump. Hold that idea — it will matter in 30 seconds.' Signal flip. Circulate silently for 3 minutes of individual writing. After 2-minute pair discussion, cold-call exactly 3 pairs using a random name stick. For each, prompt: 'Tell me your prediction AND the reasoning behind it — use the words valence band or conduction band.' After all three, say: 'We have three different predictions on the table. We will find out which model the evidence supports by the end of today.' Do NOT confirm or correct — record predictions on a sticky-note column on the whiteboard labelled 'Our Predictions — Lesson 4'. WAIT TIME: 10 seconds after each cold-call before accepting responses.	Predict-then-Investigate (Elicitation of Prior Models): By requiring learners to apply their band-gap vocabulary from Lesson 3 to a new situation before instruction, the teacher surfaces existing mental models and creates cognitive dissonance that motivates engagement with the evidence phase. Predictions are publicly recorded so learners can revisit and revise them later.	Observable indicator: Learner-written prediction cards. Teacher scans for use of band-gap vocabulary ('valence band', 'conduction band', 'energy gap'). Green flag: learner reasons that the extra electron from phosphorus may sit above the valence band and be more easily promoted. Red flag: learner says 'more atoms = better conductor' without reference to electron energy levels — flag these learners for targeted questioning in Phase 3.
Observe Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — Structured Reading & Diagram Analysis (20 min): Learners read a one-page illustrated text titled 'From Sand to Solar Panel: How Kenyan Engineers Dope Silicon' (teacher-prepared, printed). The text covers: (i) what doping means and why it is done; (ii) donor impurities (phosphorus, Group V) creating n-type silicon with free electrons as majority carriers; (iii) acceptor impurities (boron, Group III) creating p-type silicon with holes as majority carriers; (iv) a data table showing resistivity of pure Si, lightly doped Si, and heavily doped Si at 25 °C. Learners annotate diagrams using a colour code:	Before distributing the reading: 'As you read, you are a detective. Your job is to find the answer to one question: how does adding a phosphorus atom change where electrons sit in the energy band diagram? Use your coloured pens to mark every piece of evidence you find.' Circulate during reading; pause the class at the 8-minute mark and ask: 'Raise your hand if you have found the word donor — what does it mean in this context?' Cold-call 2 learners. For the data activity, say: 'Your group has 10 minutes. Plot the graph, then write exactly one sentence starting with the words: As doping concentration increases,	Evidence Annotation + Graphical Representation: Colour-coding the lattice diagrams (red = electrons, blue = holes) creates a visual–spatial encoding that makes the abstract concept of charge carriers concrete. Graphing resistivity against doping concentration moves learners from qualitative description to quantitative reasoning, mirroring the Science and Engineering Practice of Analysing and Interpreting Data (NGSS SEP 4).	Observable indicators: (1) Annotated diagrams — teacher checks that n-type diagrams show a free electron OUTSIDE a covalent bond and p-type shows a missing bond (hole). (2) Group graph sentences — listen for correct inverse relationship language ('as doping increases, resistivity decreases'). Red flag: groups who write 'as doping increases, conductivity decreases' have inverted the relationship — pull these groups aside during Phase 3 transition for a 2-minute targeted re-explanation using the data table.

	RED pen = free electrons (n-type), BLUE pen = holes (p-type). ACTIVITY B — Data Table Analysis (10 min): In groups of 4, learners receive a second data card showing resistivity vs. doping concentration for phosphorus-doped silicon (5 data rows). Groups plot the trend on a mini-graph (axes pre-drawn on the card) and write one sentence summarising the relationship. One graph per group is posted on the wall.	resistivity…' After groups post graphs, do a 90-second gallery comparison: 'Look at all four graphs. Do they all show the same trend? Nod if yes.' Address any discrepant graphs immediately by returning the group to the data table. WAIT TIME: 15 seconds after posing the graph-summary question before accepting any answer.		
Explain Phase 📹 **VIDEO:** [The periodic table - classification of elements] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Electric Charge and Electric Force (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY — 'Charge Carrier Role Play + Explanation Writing' (18 min): The class moves desks to create an open space. 10 volunteers stand in a 2×5 grid representing silicon atoms. Teacher gives 2 volunteers a red card (free electron from phosphorus doping — n-type scenario) and asks them to move freely through the grid when a 'voltage' signal (teacher clap) is given. The class observes and discusses: 'Why can the red-card electrons move but in pure silicon they could not?' Then the scenario switches to p-type: 2 volunteers hold blue cards (holes). The teacher demonstrates hole movement by having atoms pass the 'missing bond' card in the opposite direction to electron flow. After the role play, all learners return to seats and individually write a structured explanation using the sentence-stem scaffold: 'In n-type silicon, phosphorus atoms act as ______ because they ______. This means that when a voltage is applied, ______ move toward the ______ terminal, creating a current. In p-type silicon, ______ act as charge	Before the role play, set clear expectations: 'This is a scientific model — every movement you make must represent something real happening inside silicon. I will ask you to justify your movement.' During the role play, narrate for the seated audience: 'Watch the red-card learner — they are the extra electron from phosphorus. In our band model from Lesson 3, which band are they already in?' Cold-call 2 seated learners to answer. After role play, give 8 minutes for individual explanation writing. Circulate and read over shoulders. When you see a learner write a strong sentence, say: 'I am going to ask you to read that sentence to the class — be ready.' After 8 minutes, cold-call 3 learners to read their phenomenon-connection sentences. For each, ask the class: 'Do you agree? Give a thumbs-up, thumbs-sideways, or thumbs-down.' Address thumbs-sideways/down responses: 'What would you change about that sentence?' WAIT TIME: 10 seconds after 'Do you agree?' before asking for thumbs.	Embodied Cognition Role Play + Structured Writing (NGSS SEP 6 — Constructing Explanations): The physical role play externalises the abstract movement of charge carriers, making the directionality of electron vs. hole current viscerally understandable. The scaffolded sentence-stem writing then forces learners to translate the embodied experience back into precise scientific language, consolidating conceptual understanding and building towards the written explanations required in the final lesson.	Observable indicators: (1) During role play — do volunteers move red-card electrons toward the positive terminal? If they move the wrong way, pause and ask the class: 'Which terminal should a negative charge move toward?' (2) Written explanations — teacher checks for: correct identification of electrons as majority carriers in n-type; correct identification of holes as majority carriers in p-type; a phenomenon-connection sentence that mentions the p-n junction or doping in the context of the solar panel. Green flag: learner writes 'the solar panel uses both n-type and p-type silicon joined together so that current can only flow in one direction.' Red flag: learner writes 'doping just makes silicon conduct like a metal' — schedule a 3-minute one-on-one conversation at the end of the lesson.

	carriers because ______.' Learners then connect to the phenomenon: 'The silicon in the Kajiado solar panel is not pure — it must be doped. Explain in two sentences why this matters for generating electricity.'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes — one green, one orange. On the GREEN note, they write one thing they now understand about the Kajiado solar panel that they did not understand at the start of Lesson 4 (must include the word 'doping', 'n-type', or 'p-type'). On the ORANGE note, they write one new question that Lesson 4 has raised for them — something they still cannot explain about the panel's behaviour. Learners post both notes on the class Driving Question Board in the designated 'Lesson 4 Evidence' column. The teacher reads 5–6 notes aloud and groups them into emerging themes: (a) questions about what happens at the boundary between n-type and p-type silicon, (b) questions about why temperature still matters if the silicon is already doped, (c) questions about how the solar panel converts light — not just heat — into electricity.	Say: 'Our Driving Question Board is the story of our growing understanding. Green notes show what we have figured out — orange notes show where our model still has gaps.' Give learners 4 minutes to write both notes before posting. After posting, spend 5 minutes reading selected notes aloud. For each orange question that touches on the p-n junction, say: 'Hold that question — write it in your notebook and underline it. That is exactly what we will investigate in Lesson 5 using a computer simulation.' For any orange question about light absorption (photons), say: 'Excellent — that question goes even deeper than today's lesson. We will park it here and return to it.' Point to the parking-lot section of the DQB. Do NOT resolve orange questions — preserve curiosity. WAIT TIME: After reading each orange note, pause 10 seconds and ask: 'Does anyone want to add to this question or make it more specific?'	Public Knowledge-Building (DQB Protocol): The two-colour sticky-note system makes individual sense-making visible and collective. By physically sorting orange questions into thematic clusters, the teacher models the scientific practice of identifying productive lines of inquiry (NGSS SEP 1 — Asking Questions), and creates a bridge to the next lesson's investigation agenda.	Observable indicators: Green notes — teacher checks that learners are making specific claims ('I now understand that phosphorus gives silicon a free electron that can carry current') rather than vague ones ('I understand doping better'). Orange notes — teacher checks that questions are phenomenon-connected (about the solar panel, about temperature, about the junction) rather than off-topic. A high proportion of orange questions about the p-n junction boundary signals readiness for Lesson 5. Flag learners with no orange note — they may have surface-level understanding and need probing questioning.
Model Building Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners retrieve their cumulative 'Silicon Solar Panel Explanatory Model' sheet begun in Lesson 1 and updated in each subsequent lesson. Today they add a new section labelled 'Lesson 4 — Doping'. Using a pre-drawn template on the model sheet showing two adjacent blocks of silicon (left block = n-type, right	Distribute model sheets and say: 'Your model should now tell the story from Lesson 1 all the way to today. A stranger who has never studied physics should be able to look at your model and understand why silicon — not copper, not rubber — is the right material for a solar panel.' Give 10 minutes for individual model completion, then	Cumulative Explanatory Model Revision (NGSS SEP 2 — Developing and Using Models): By returning to and visibly extending the same model sheet across all 7 lessons, learners experience science as a progressive, evidence-driven process of model refinement. The social comparison step (group best-model	Observable indicators: (1) Correct directionality of charge carrier arrows (electrons in n-type move toward positive terminal; holes in p-type move toward negative terminal — i.e. in opposite directions). (2) Correct placement of donor (+) and acceptor (−) ion symbols. (3) Caption that explicitly names the solar panel context.

for similar videos 📖 **HTML:** Electric Charge and Electric Force (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	block = p-type), learners: (a) populate the n-type block with symbols for free electrons (red dots) and phosphorus donor atoms (+); (b) populate the p-type block with symbols for holes (blue circles) and boron acceptor atoms (−); (c) draw arrows showing the direction majority charge carriers would move in each block if a voltage were applied; (d) write a caption at the bottom: 'This is the basic structure inside every silicon solar panel cell in Kajiado — and in every mobile phone chip used to run M-PESA across Kenya.' Groups of 4 compare their models and agree on one 'best model' to display. Teacher photographs all displayed models for inclusion in the next lesson's review.	5 minutes for group comparison. Circulate and ask probing questions: 'Why did you draw the arrows in that direction?', 'What would happen to the holes if you reversed the voltage?', 'Which side of this junction do you think faces the sun — and does it matter?' Cold-call 2 groups to present their model to the class in 60 seconds each. After each presentation, ask the class: 'What does this group's model explain that yours does not — or vice versa?' WAIT TIME: 15 seconds after each model presentation question before accepting responses.	negotiation) requires learners to justify their representational choices, deepening conceptual ownership. Connecting the model caption to M-PESA and Kajiado irrigation grounds the abstraction in lived Kenyan experience.	Red flag: learner draws electrons moving in both n-type and p-type blocks — this indicates confusion between intrinsic and extrinsic semiconductors and requires a targeted correction using the band-gap diagram before Lesson 5.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PREDICTION ACCURACY — What fraction of learners correctly predicted that the phosphorus extra electron would be more mobile than a covalent-bond electron? If fewer than 40% predicted correctly, consider adding a 5-minute band-gap review at the start of Lesson 5. (2) ROLE PLAY EFFECTIVENESS — Did the hole-movement demonstration (passing the blue card opposite to electron direction) create genuine understanding or confusion? If learners in their model-building phase drew holes moving in the same direction as electrons, the directionality concept needs to be revisited via the PhET simulation in Lesson 5 with explicit attention to current direction. (3) DQB ORANGE NOTES — Count how many orange notes specifically mention the p-n junction boundary. If fewer than half the class generated a junction question, the transition to Lesson 5 may need a more explicit bridging prompt: show the two model blocks and ask 'what happens right at the line where n-type meets p-type?' (4) PHENOMENON CONNECTION — Did learners' explanation writing and model captions explicitly name the Kajiado solar panel? If many learners wrote generic statements about semiconductors without connecting to the phenomenon, use the first 5 minutes of Lesson 5 to re-anchor by re-showing the Kajiado farmer data from Lesson 1. (5) EQUITY CHECK — Were the same learners cold-called repeatedly? Review name-stick records and ensure all learners were reached across Lessons 1–4. Prioritise learners who have not yet been cold-called for the first three questions in Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Colour-annotated silicon lattice diagrams showing phosphorus (donor) atoms with a free red-dot electron and boron (acceptor) atoms with a blue-circle hole; a plotted graph showing that resistivity decreases as doping concentration increases in phosphorus-doped silicon.
What did I learn?	Doping is the deliberate addition of impurity atoms to pure silicon to control its conductivity. Phosphorus (Group V) donates a free electron, creating n-type silicon where electrons are the majority charge carriers. Boron (Group III) creates a hole (missing electron), producing p-type silicon where holes are the majority charge carriers. The higher the doping concentration, the lower the

	resistivity — making doped silicon far more conductive than pure (intrinsic) silicon.
How does this explain the phenomenon?	The silicon in the Kajiado solar panel is not pure — it is a precisely engineered combination of n-type and p-type silicon. This doping is what allows the panel to conduct electricity in a controlled direction. In Lesson 5 we will investigate what happens at the boundary between the n-type and p-type regions (the p-n junction) to explain why current flows only one way — and why high afternoon temperatures in Kajiado reduce the panel's power output.

LESSON 5 (40 minutes): PhET Semiconductor Lab: Exploring p-n Junctions, Bias, and Temperature Effects	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To use a PhET simulation as a systematic investigation tool, collecting I-V data for p-n junctions under varied doping and temperature conditions, and using the evidence to construct an explanation for why the Kajiado solar panel produces less power in hot afternoon conditions.
Knowledge	Learners will be able to: (1) describe how a p-n junction is formed when p-type and n-type silicon are joined; (2) explain the role of the depletion region in controlling current flow; (3) distinguish between forward bias and reverse bias behaviour using I-V graphs; (4) explain how increasing temperature increases minority carrier generation in doped silicon, reducing the built-in voltage of the junction and degrading solar panel efficiency.
Skills	Learners will systematically vary independent variables (doping type, doping concentration, temperature, applied voltage) in the PhET simulation, record quantitative I-V data in tables, plot and interpret I-V characteristic curves, and construct evidence-based written explanations linking simulation data to the anchoring phenomenon.
Attitudes	Learners will demonstrate scientific curiosity by interrogating unexpected simulation results, intellectual honesty by recording data that may contradict initial predictions, and appreciation for how simulation tools extend investigative possibilities beyond what physical lab equipment in a Kenyan school can provide.
Key Inquiry Question	How do the electrical properties of materials depend on their atomic and energy band structure? (KIQ 1) — addressed by linking p-n junction behaviour directly to the band diagrams and doping models constructed in Lessons 3 and 4.
Purpose in Storyline	This lesson is the pivotal evidence-gathering session in the sequence. Learners have the conceptual tools from Lessons 1-4 (band theory, intrinsic carriers, n-type and p-type doping) and now use the PhET simulation to generate their own quantitative evidence about how junctions and temperature interact. This positions them to give a complete, model-based explanation of the anchoring phenomenon before the applied case studies in Lesson 7.
Safety Notes	No physical hazards. Computers or tablets must be handled with care. If using a school computer lab, learners should not change system settings. Teacher should verify PhET Semiconductor simulation is accessible (browser-based at phet.colorado.edu or pre-downloaded) before the lesson begins. Inform learners that simulation results represent idealised models and real silicon p-n junctions may show slight deviations.

B. LESSON OVERVIEW
Learners open the lesson by revisiting a one-sentence prediction about the Kajiado solar panel written at the end of Lesson 4. They then spend the core of the lesson in structured PhET Semiconductor simulation investigation, working in pairs to vary doping type, doping concentration, temperature, and bias voltage while recording I-V data. Three targeted investigations are sequenced: (A) junction formation and depletion region visualisation, (B) forward versus reverse bias I-V curves, and (C) temperature effects on junction voltage and current. In the Explain phase, pairs share findings and co-construct a class I-V characteristic poster. The lesson closes with learners updating the Driving Question Board with new evidence statements and drafting a revised annotated model of why high temperature reduces the solar panel output — directly advancing the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 4 exit-ticket card (or a printed half-sheet provided by the teacher) on which they wrote a prediction: 'At high temperature, the Kajiado solar panel produces less power because...' They share predictions with their neighbour and identify the ONE thing they are most uncertain about. Class predictions are collected verbally — teacher records three to four distinct ideas on the board under the heading 'Our Current Best Explanations'. Learners then read a two-paragraph stimulus about how a p-n junction is the heart of every solar cell — describing that when p-type and n-type silicon are placed in contact, electrons from the n-side and holes from the p-side diffuse across the boundary and recombine, leaving behind a thin depletion region with a built-in electric field. Learners predict: (1) What will happen to current if we apply a voltage that pushes electrons toward the junction (forward bias)? (2) What will happen if we apply a voltage that pulls electrons away (reverse bias)? (3) If temperature increases inside the panel, will the built-in voltage of the depletion region increase or decrease — and why? Each pair records three predictions in their investigation booklet before touching the simulation.	Begin by asking: 'Look back at the prediction you wrote at the end of last lesson. Has anything changed in your thinking since then? Share with your partner for 30 seconds.' WAIT 30 seconds. Then cold-call three pairs: 'Akinyi, what is one thing your partner said that was different from your own idea?' After recording predictions, distribute or display the stimulus paragraph. Say: 'Before we open the simulation, I want you to commit to three specific predictions — write them down so we can test them. A prediction you cannot test is not useful in science.' Circulate and check that predictions are quantitative where possible (for example: 'current will increase' rather than just 'something will happen'). Prompt deeper thinking: 'You said current will increase in forward bias — increase by a little or a lot? What does your band diagram from Lesson 4 suggest about the energy barrier the electrons must overcome?' WAIT 20 seconds for pairs to discuss. Do not confirm or deny predictions — preserve the epistemic uncertainty that will motivate careful observation.	Prediction-before-observation activates prior knowledge from Lessons 3 and 4 and creates cognitive dissonance points that the simulation data will resolve. Recording predictions publicly (on the board) makes the class accountable to evidence and models scientific practice. The three-prediction structure (forward bias, reverse bias, temperature effect) maps directly onto the three simulation investigations, creating a coherent predict-observe-explain arc.	Teacher scans written predictions during circulation. Look for: (a) learners who predict symmetrical I-V curves for forward and reverse bias — this reveals a misconception about junction rectification that the simulation will directly address; (b) learners who predict higher temperature increases efficiency — this is the key target misconception linked to the anchoring phenomenon; (c) learners who cannot connect their prediction to the band model — prompt them to redraw the n-type and p-type band diagram side by side before proceeding. Flag these pairs for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** The periodic table - classification of	Learners work in pairs at a computer or tablet running the PhET Semiconductor simulation (University of Colorado Boulder). The teacher displays a structured	As learners begin Investigation A, circulate and ask: 'Point to the depletion region on your screen. What charge carriers are missing from that region, and why?' WAIT	The three-investigation sequence uses a productive scaffolding structure: Investigation A builds spatial understanding of the junction, Investigation B generates	Teacher checks results tables during circulation for: (a) correct identification of rectification (asymmetric I-V curve) — if absent, prompt the pair to

elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	investigation guide with three sequential tasks, each taking approximately 5 to 6 minutes. INVESTIGATION A — Junction Formation (5 min): Learners select n-type silicon on one side and p-type on the other. They observe the depletion region forming automatically, note the direction of the built-in electric field arrow, and sketch the energy band diagram showing the Fermi level alignment across the junction. They record: width of depletion region (relative, in simulation units), direction of built-in field, and which side has positive fixed charges versus negative fixed charges. INVESTIGATION B — I-V Characteristics (8 min): Learners apply bias voltage from -2V to +2V in 0.25V steps, recording current at each step in a results table. They plot current (y-axis) versus voltage (x-axis) on pre-printed graph axes in their booklet. They annotate two regions: forward bias (low resistance, large current) and reverse bias (high resistance, near-zero current until breakdown). They note the threshold voltage at which current rises sharply in forward bias and relate it to the height of the energy barrier in the depletion region. INVESTIGATION C — Temperature Effect (7 min): Learners keep bias voltage fixed at +0.5V (forward bias) and vary temperature from 200K to 400K in steps of 50K, recording current and noting any change in the visualised depletion region width. They also record the built-in voltage displayed by the simulation at each temperature step. Key observation: as temperature rises, minority carrier concentration	15 seconds. For Investigation B, pause the class after 4 minutes and say: 'Raise your hand if your I-V curve looks the same in both directions — symmetric. Keep your hand up if you predicted it would be symmetric.' Acknowledge raised hands without judgment, then say: 'Keep collecting data. We will come back to this.' This sustains productive tension. During Investigation C, target the pairs flagged in the Predict phase. Ask them directly: 'You predicted higher temperature would make the panel more efficient. What are you actually seeing happen to the built-in voltage as temperature goes up? Read the number off the screen and tell me.' WAIT 20 seconds. If a pair is confused about why depletion region width changes with temperature, prompt: 'Think about what doping concentration does from Lesson 4 — temperature is doing something similar by freeing more carriers. What does more free carriers do to the depletion region?' Do not give the answer — use Socratic probing. At 18 minutes into the lesson, call time and say: 'Everyone freeze your simulation. Before we discuss, write one sentence summarising the single most surprising thing you observed today.'	the quantitative I-V evidence, and Investigation C connects directly to the anchoring phenomenon. The structured results table prevents data overload and ensures all pairs have comparable evidence for the class discussion. The 'most surprising observation' prompt at the end of the Observe phase is a cognitive consolidation moment that prepares learners for the Explain phase.	compare current at +1V and -1V explicitly; (b) correct direction of built-in field (pointing from n-side to p-side, opposing diffusion current) — if wrong, ask the learner to identify where positive ions are located; (c) in Investigation C, whether the learner records decreasing built-in voltage with increasing temperature — this is the key quantitative evidence for the anchoring phenomenon explanation. A quick show-of-thumbs check after Investigation B ('Thumbs up if your graph shows much larger current in one direction') gives the teacher a rapid class-wide read before moving to Explain.

	increases (more thermally generated electron-hole pairs), the built-in voltage decreases, and — critically — the open-circuit voltage a solar cell can deliver falls. Learners note this in their booklet with the heading 'What temperature does to the junction.'			
Explain Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Pairs join into groups of four to compare I-V graphs and reconcile any differences in their data. Each group is assigned one of three explanation tasks: Group Type 1 explains why the I-V curve is asymmetric (rectification) using the depletion region and energy barrier model. Group Type 2 explains the threshold voltage in forward bias in terms of overcoming the built-in potential barrier, using the band diagram. Group Type 3 explains why increasing temperature from 200K to 400K decreases the built-in voltage and therefore reduces the maximum voltage a solar cell can deliver — connecting directly to the Kajiado anchoring phenomenon. Each group constructs a mini-poster (A3 paper or whiteboard section) with: (a) a labelled diagram or graph from their simulation data, (b) a band-theory explanation in their own words, and (c) one sentence explicitly connecting their finding to the Kajiado solar panel. Groups display posters and give a 90-second verbal summary to the class. Teacher facilitates a brief whole-class synthesis: 'So what have we now established as evidence about why hot afternoon sun produces less power?' Learners co-construct a class consensus statement recorded on the board: 'At high temperatures,	During group work, circulate and push for precision. For Group Type 1: 'You have said current flows easily in forward bias. But what is physically happening to the energy barrier? Use the words barrier height in your explanation.' For Group Type 2: 'Your threshold voltage is about 0.6 to 0.7V in the simulation. Where does that number come from? Look at your band diagram — what energy gap are electrons overcoming at the junction?' For Group Type 3 — the most critical group for the anchoring phenomenon: 'Your data shows built-in voltage drops from about 0.9V at 200K to about 0.5V at 400K. Now Kajiado afternoon temperature is about 35 degrees Celsius — that is 308K. What does your graph predict will happen to the panel voltage on that hot afternoon versus a cool 18 degrees Celsius morning at 291K? Can you put a number on it?' WAIT 25 seconds. During poster presentations, use probing questions: 'Wanjiku says the junction voltage drops with temperature. Mwangi, do you agree? What evidence from your own simulation supports or challenges that?' After all presentations, facilitate the consensus statement using think-pair-share: 'In one sentence, what is our best evidence-based explanation for the anchoring	The jigsaw-style task assignment ensures that three complementary explanations (rectification, threshold voltage, temperature effect) are developed in parallel and then integrated in the class synthesis. This mirrors how scientific knowledge is built from multiple lines of evidence. Constructing a shared consensus statement makes the class explanation public and accountable, and it directly advances the driving question. The explicit connection to Kajiado in each poster task ensures the phenomenon remains the anchor, not an afterthought.	Evaluate posters using three criteria: (1) Is the diagram or graph correctly labelled with physics terms? (2) Does the explanation invoke the energy band model or depletion region — not just surface-level description? (3) Is the connection to the Kajiado phenomenon specific and accurate? Learners who produce vague connections (for example: 'heat makes it worse') without a mechanism should be prompted to add a causal chain. The teacher can use a quick whole-class exit check: 'On your mini-whiteboard or scrap paper, draw the I-V curve for a p-n junction and mark forward and reverse bias regions. This should take 60 seconds.' Scan the room for the characteristic rectifier shape before dismissing to the DQB phase.

	thermally generated minority carriers increase, reducing the p-n junction built-in voltage, which lowers the open-circuit voltage of the solar cell and therefore reduces the electrical power output — even though more photons are arriving at the panel.'	phenomenon right now? Write it, share it with your partner, then we will build one class sentence together.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board — a large display that has been maintained since Lesson 1. Each pair receives two sticky notes. On the first (green): they write one new piece of evidence they have collected today that helps answer the driving question. On the second (orange): they write one new question that their simulation results have raised — something they still cannot explain or are curious about. Examples of expected green evidence notes: 'At 400K the built-in voltage dropped to 0.5V meaning the solar cell gives less output voltage even with bright light', 'Reverse bias blocks nearly all current — that is why a solar cell only works in one direction'. Examples of expected orange question notes: 'If we cool the solar panel with water irrigation spray would it produce more power?', 'What happens inside the solar cell when a photon hits — we only studied voltage bias not light'. Learners place their notes on the DQB under the sub-headings 'New Evidence' and 'New Questions'. Teacher reads three to four notes aloud, acknowledging the quality of evidence statements and validating the new questions as important. The teacher adds one teacher-generated note: 'We now have a physical mechanism to explain the Kajiado phenomenon	Say: 'The DQB is a living record of our thinking. Look at what we put up in Lesson 1 — can you find a question from that lesson that you can now answer with evidence from today?' WAIT 30 seconds and cold-call two learners to respond. Then direct attention to today's new additions: 'I am seeing some excellent evidence notes. Ochieng has written that the built-in voltage decreased from 0.9 to 0.5V over a 200K temperature range — that is precise, quantitative, and directly connected to our phenomenon. That is exactly the quality of evidence statement a scientist would write.' Highlight the gap explicitly: 'Notice that we now understand why high temperature reduces voltage output. But our DQB still has a question from Lesson 1: how does the solar panel turn light into electricity in the first place? Keep that question in mind — it will be central in Lesson 7.' Do not over-answer new questions; preserve curiosity. If a learner raises an unexpected and sophisticated question (for example, about photon energy and band gap), say: 'That is a genuinely important question. Write it on an orange note and put it up — it deserves a full answer, not a rushed one right now.'	The DQB update is a metacognitive consolidation strategy that makes learning progress visible. By physically moving sticky notes or adding new ones, learners experience the iterative nature of scientific inquiry — knowledge grows but also generates new questions. The teacher modelling how to read a DQB question and connect it to today's evidence demonstrates the disciplinary practice of using evidence to resolve questions progressively. Leaving one question explicitly open ('how does a photon create the electron-hole pair') sustains narrative momentum into Lessons 6 and 7.	Teacher reads all sticky notes after the lesson to assess: (a) Are evidence notes causal and specific (invoking band theory, junction voltage, or I-V data) or superficial? (b) Do new questions reveal genuine curiosity and engagement with the model, or confusion about previously covered content? Notes showing persistent confusion about n-type versus p-type distinctions indicate the Lesson 4 content needs brief revisiting at the start of Lesson 6. Collect the most insightful orange question notes as diagnostic data to inform the design of the Lesson 7 case study discussions.

	— but we have not yet explained how a photon creates the initial electron-hole pair that starts the current. That is a gap we will address in Lesson 7.'			
Model Building Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, each learner individually updates their personal annotated model diagram — a visual they have been developing since Lesson 1. Today they add two new elements: (1) A p-n junction diagram showing the depletion region, built-in field direction, and the asymmetric I-V curve shape (sketched, not plotted in full). (2) A temperature annotation box connected by an arrow to the junction diagram, with the statement: 'Higher temperature generates more thermal electron-hole pairs, reducing the built-in voltage of the junction, lowering the open-circuit voltage of the solar cell, and reducing power output — even when photon input is high. This explains the Kajiado afternoon power drop.' Learners also write one revision sentence at the bottom of their model: 'My model has changed from Lesson 4 because...' Expected completion: learners should note that they previously understood doping creates free carriers, but now understand that the junction formed between p and n regions creates a directional electric field whose strength is temperature-dependent — and that this junction voltage is what gets eroded by afternoon heat in Kajiado.	Say: 'Your model should now be more powerful than it was at the start of this lesson. In Lesson 1 you had no way to explain why the solar panel worked less well in the afternoon. Write the revision sentence and be specific — name the physics mechanism that you now understand that you did not before.' WAIT 3 minutes for individual silent writing. Circulate and read over shoulders — do not interrupt unless a learner is adding a factually incorrect statement to their model. If a learner writes only 'because of temperature' without a mechanism, kneel to their level and say quietly: 'That tells me what changes, but not why. Add the word junction voltage to your sentence — where does that fit in your causal chain?' Collect models at the door as an exit ticket — scan for the presence of: p-n junction diagram, depletion region label, asymmetric I-V curve sketch, and the causal temperature sentence. Return models at the start of Lesson 6 with one written feedback comment per learner.	The personal annotated model is the cumulative sensemaking artefact of the entire unit. Requiring a 'my model has changed because' sentence forces learners to engage in explicit knowledge revision — a high-order metacognitive practice aligned with CBE competency development. By collecting models as exit tickets, the teacher gains individual-level formative data that informs differentiated support in Lesson 6, particularly for learners who are still conflating temperature effects in semiconductors with temperature effects in metals (a persistent cross-cutting misconception from everyday experience with conductors).	Exit ticket model diagrams are assessed against four criteria: (1) p-n junction drawn with correct labelling of n-side, p-side, and depletion region; (2) built-in field direction shown correctly (n to p is positive ion layer, field points from n to p interior); (3) I-V curve sketch shows rectification (large current in one direction, small in the other); (4) temperature annotation correctly states that higher temperature reduces built-in junction voltage and therefore reduces solar cell output voltage. Any model missing criterion 4 — the critical link to the anchoring phenomenon — is flagged for a brief one-to-one clarification conversation at the start of Lesson 6 before the superconductivity topic is introduced.

D. TEACHER REFLECTION

After the lesson, consider: Did all pairs successfully complete all three PhET investigations within the time available, or did Investigation C get rushed? If time was short,

Investigation C (temperature effect) is the most critical for the anchoring phenomenon — it should be protected even if Investigation A (junction formation visualisation) is abbreviated. Were learners able to move fluidly between the simulation data and the band theory model from Lessons 3 and 4, or did they treat the simulation as a black box? If band theory connections were weak, open Lesson 6 with a 3-minute retrieval quiz on n-type and p-type band diagrams before introducing superconductors. Were the DQB orange (new question) notes genuinely curious and connected to the model, or were they surface-level? High-quality new questions (for example, about photon energy thresholds, or cooling solar panels with water) indicate deep engagement and should be named and celebrated at the start of Lesson 6. Consider whether access to the PhET simulation was equitable — if some pairs shared one device between three or four learners, plan a brief repeat of Investigation C as a whole-class teacher-led demonstration at the start of Lesson 6 to ensure all learners have the evidence they need for the Lesson 7 case studies.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the PhET simulation, we observed that a p-n junction allows large current in one direction (forward bias) and blocks current in the other (reverse bias), producing an asymmetric I-V curve. We also observed that as temperature increased from 200K to 400K, the built-in voltage of the junction decreased and minority carrier current increased, consistent with greater thermal generation of electron-hole pairs in the doped silicon.
What did I learn?	We learned that a p-n junction forms a depletion region with a built-in electric field that acts as a directional barrier to charge flow. In forward bias, the applied voltage reduces this barrier height, allowing exponentially increasing current. In reverse bias, the barrier is increased, blocking current. Temperature erodes the built-in voltage because more thermally generated minority carriers partially neutralise the depletion region charge, reducing the internal field strength.
How does this explain the phenomenon?	We can now explain the anchoring phenomenon: on a hot Kajiado afternoon (35 degrees Celsius, approximately 308K), thermal energy generates additional electron-hole pairs in the p-n junctions of the solar panel, reducing the built-in depletion region voltage. This lowers the open-circuit voltage the panel can deliver, decreasing electrical power output — even though the number of photons hitting the panel is higher than on a cool morning. The semiconductor junction, not just the light intensity, determines how much power is produced.

LESSON 6 (80 minutes): Case Studies — Solar Panels, Microchips & Superconducting Magnets: Applying Semiconductor Understanding to Kenyan Contexts

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate and apply cumulative understanding of conductors, insulators, and semiconductors by analysing three structured Kenyan-context case studies — rural solar panels in Turkana, microchips in M-PESA handsets, and superconducting MRI magnets at Kenyatta National Hospital — thereby constructing a full explanatory model for the anchoring phenomenon.
Knowledge	Learners explain how doping of silicon creates n-type and p-type semiconductors, how a p-n junction converts light into electrical current in a photovoltaic cell, how temperature affects carrier mobility and efficiency, and how extreme cooling produces superconductivity. They connect band theory to each real-world device.
Skills	Learners apply evidence-gathering, data interpretation, structured argumentation (Claim–Evidence–Reasoning), collaborative group analysis, and annotated model construction to evaluate how semiconductor properties underpin three distinct technologies.
Attitudes	Learners appreciate that scientific understanding of atomic-scale material properties has transformed everyday life for Kenyan communities — from powering rural schools to enabling mobile money to supporting medical diagnosis — and reflect on equitable access to these technologies.
Key Inquiry Question	How has the understanding and engineering of semiconductor materials transformed modern technology and everyday life in Kenya?
Purpose in Storyline	Lesson 6 is the penultimate evidence-gathering lesson. By this point learners have built understanding of band theory, doping, p-n junctions, and temperature effects on semiconductors. In this lesson they synthesise all prior evidence by applying it to three rich Kenyan case studies. This prepares them to fully answer the driving question in Lesson 7, where the DQB is finalised and the complete annotated model is produced.
Safety Notes	No hazardous materials are used in this lesson. If physical solar panels or electronic components are handled during the case study activity, learners must not connect any component to mains electricity. Remind learners that MRI magnets involve extremely strong magnetic fields — no metallic objects near any demonstration materials. Standard classroom conduct rules apply throughout group work.

B. LESSON OVERVIEW

Learners work in three expert groups, each analysing one Kenyan-context case study: (1) silicon photovoltaic panels powering a school in Turkana County and why output drops on hot afternoons, (2) microchips inside M-PESA mobile handsets and how doped semiconductors enable logic switching at microscopic scales, and (3) superconducting electromagnets inside the MRI scanner at Kenyatta National Hospital and how near-absolute-zero cooling eliminates resistance entirely. Each group uses a structured Case Study Analysis Sheet (Claim–Evidence–Reasoning format), draws on all class posters, summary tables, and the YouTube semiconductor video revisited at the start. Groups then jigsaw to teach one another, and the class co-constructs a unified annotated model on the board that directly addresses the driving question: why does the Kajiado solar panel produce less power on a hot afternoon? The DQB is updated with final refinements, setting up the full resolution in Lesson 7.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Threshold 8: The Modern Revolution] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a single slide (or hand-drawn board sketch) displaying three images side-by-side: a solar panel on a school rooftop in Lodwar, Turkana County; a person using a basic smartphone to send M-PESA money in Kisumu market; and a patient entering an MRI scanner at Kenyatta National Hospital, Nairobi. Below the images is the prompt: 'All three technologies depend on the same family of materials you have been studying. Before we dig in — write down: (a) which semiconductor property or concept from our lessons do you think is MOST important for each device, and (b) what question about each device you still cannot fully answer.' Learners write individually for 4 minutes in their science notebooks, then share predictions with a shoulder partner for 2 minutes. The teacher cold-calls 6 learners (2 per device image) to voice their predictions aloud. Predictions are recorded in three columns on the board under each image heading. These predictions remain visible throughout the lesson and are revisited during Phase 4.	Display the three-image slide and say: 'We have spent five lessons gathering evidence about how materials conduct — or refuse to conduct — electricity, and why silicon is special. Today, before I tell you anything new, I want you to commit to a prediction.' Allow 4 minutes of silent individual writing — enforce silence, no devices. After 4 minutes say: 'Turn to your shoulder partner, share your two answers — you have 2 minutes.' After partner talk, cold-call exactly 6 learners using the class register (learner 3, 9, 15, 21, 27, and 33 — or equivalent spacing). For each response, probe with: 'What evidence from our previous lessons supports that idea?' Allow 10 seconds of WAIT TIME after each probe before accepting a response. Record key terms from responses in the three columns on the board. Do NOT correct misconceptions yet — note them visibly with a question mark (e.g., 'Wambua says doping makes the panel hotter — ❓') so they can be resolved during sensemaking.	Activation of Prior Knowledge via Prediction Columns — by committing predictions to paper and then hearing peers' varied ideas displayed on the board, learners activate their full five-lesson knowledge base and surface residual misconceptions. The three visible columns create a cognitive scaffold: learners will mentally 'check off' each column as evidence is presented during the case study work, reinforcing retrieval and integration.	Observable indicator: each learner's notebook contains at least two written predictions (one semiconductor concept per device, one unanswered question per device) before partner talk begins. The teacher circulates during the 4-minute writing phase and notes which learners leave spaces blank — these learners are prioritised for targeted questioning during Phase 3. Red flag: if more than one-third of cold-called learners cannot name any semiconductor concept for a device, the teacher pauses to do a rapid 2-minute whole-class oral recap of the band theory summary poster before proceeding.
Observe Phase 📹 **VIDEO:** [Threshold 8: The Modern Revolution] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)]	The class is divided into three expert groups (Group A: Solar Panels / Turkana; Group B: Microchips / M-PESA; Group C: Superconductors / KNH MRI). Each group receives a printed or tablet-displayed Case Study Evidence Pack containing: (i) a one-page structured reading written at Grade 10 level in clear English, (ii) a data table or diagram specific to their device (Group A: a graph of solar panel power output vs. panel temperature from 15 °C	Before releasing groups, say: 'Your job in the next 15 minutes is to be the expert on your case study. Your group will later teach the rest of the class — so everyone in the group must understand the evidence, not just the fastest reader.' Assign roles within groups aloud: Reader, Recorder, Diagram Annotator, and Timekeeper. Circulate between groups every 3–4 minutes. At Group A, prompt: 'Look at the power-vs-temperature graph —	Jigsaw Expert Group Reading with Annotated Data — each group becomes the class authority on one case study. The structured reading plus data visualisation (graph or diagram) grounds their analysis in concrete, Kenya-specific evidence rather than abstract theory. By linking new evidence explicitly back to class posters from prior lessons, learners build a connected knowledge structure rather than treating each lesson as isolated	Observable indicator: every group's Case Study Analysis Sheet has at least three evidence statements in row one and at least two explicit links to prior lesson concepts in row two by the end of the 15-minute window. The teacher collects a quick visual check by asking Recorders to hold up their sheets — if fewer than two prior-lesson links appear, the teacher stops the group and asks: 'Which poster on the wall connects to your evidence? Point to it.' This

Source: CK-12 🔍 Search ARES for similar readings	to 45 °C for a silicon PV panel in Lodwar; Group B: a simplified diagram of an n-type/p-type junction used as a transistor switch inside a mobile processor, with annotated electron flow; Group C: a resistance-vs-temperature graph for a superconducting niobium-titanium alloy showing the sharp drop to zero resistance at ~9 K, with a note that MRI magnets at KNH operate at 4 K using liquid helium cooling). Groups also have access to all class posters and summary tables produced in Lessons 1–5, plus the YouTube semiconductor video (bookmarked at the relevant timestamp). Groups read, discuss, and complete the first two rows of their Case Study Analysis Sheet: 'What did we OBSERVE (evidence)?' and 'What does this REMIND US OF from our previous lessons?' — 15 minutes of structured group work.	what pattern do you see, and does it remind you of what we said about carrier mobility last lesson?' At Group B, prompt: 'How is this transistor junction different from a simple copper wire, and what would happen to the M-PESA signal if there were no doping?' At Group C, prompt: 'The resistance drops to exactly zero — is that what we predicted from band theory alone? What is different about this material below 9 K?' Use cold-call within groups: call on the quietest member by name and ask them to read one key sentence from the evidence pack. Allow 10 seconds of WAIT TIME. With 3 minutes remaining, give a time check: 'Three minutes — make sure your Recorder has written at least three pieces of evidence in row one of your sheet.'	content.	anchors the group before jigsaw sharing.
Explain Phase 📹 **VIDEO:** Threshold 8: The Modern Revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups now complete their Case Study Analysis Sheets fully, adding rows three and four: 'What CLAIM can we make about how semiconductor properties explain this technology?' and 'What is our REASONING — which specific property (band gap, doping, p-n junction, temperature effect, superconductivity) is doing the work and why?' (10 minutes within expert groups.) Then the class jigsaws: new mixed groups of three are formed, one member from each original expert group per new group. Each expert teaches their case study to the other two members using their completed sheet and pointing to relevant class posters (8 minutes). After jigsaw, the whole class reconvenes. The teacher facilitates	For the Claim–Evidence–Reasoning completion phase, circulate and prompt: 'Your claim should be one sentence that a farmer in Turkana / a banker using M-PESA / a doctor at KNH could understand. Write it in plain language first, then add the physics.' During jigsaw teaching, enforce the rule: 'The expert must point to at least one class poster while explaining — no poster reference, no credit.' During whole-class discussion, cold-call at least 8 different learners across all three original groups to contribute to the three-column board table. After each contribution, ask the class: 'Does everyone agree? Thumbs up, thumbs sideways, or thumbs down — silent vote.' For any thumbs-down or sideways, ask	Claim–Evidence–Reasoning (CER) Framework with Jigsaw Teach-Back — CER makes scientific argumentation explicit and structured. The jigsaw forces every learner to deeply process one case study AND then teach it, requiring them to translate technical understanding into clear language — the highest level of Bloom's taxonomy. The three-column board synthesis creates a visual knowledge map that makes the connection between disparate technologies and the original phenomenon explicit and memorable.	Observable indicator: each group's completed CER sheet names a specific semiconductor property (not just 'electronics' or 'science') in the Claim row, cites at least one piece of data from the Evidence Pack, and the Reasoning row explicitly references band structure, doping type, or temperature effect. During jigsaw, the teacher listens for whether experts can answer clarifying questions from their peers without reading verbatim from the sheet — this reveals depth of understanding. During the whole-class discussion, the teacher notes which learners produced a complete written answer to the Kajiado prompt in their notebooks — these are reviewed at the end of the lesson.

	a 12-minute whole-class structured discussion to build a unified explanation on the board. The teacher draws a three-column table: Device | Key Semiconductor Property | Link to Kajiado Phenomenon. Learners contribute to filling in each row. The final row is left blank: 'The Kajiado Solar Panel — full explanation' — learners attempt to fill this in collaboratively, drawing on all three case studies. The teacher scribes learner language, only refining vocabulary when necessary.	that learner: 'Tell us what you would change and why' — allow 10 seconds of WAIT TIME before they respond. For the final blank row (Kajiado Solar Panel), say: 'This is the question we started with seven lessons ago. Using everything in this table — Group A, B, and C — can someone now give me a complete answer? You have 30 seconds to write it in your notebook first.' Cold-call 2 learners after 30 seconds. Scribe their combined explanation on the board. Ask the class: 'Is anything missing? What would make this explanation stronger?' Allow 10 seconds of WAIT TIME.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Threshold 8: The Modern Revolution](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board that has been built across all six lessons. Each learner receives two sticky notes of different colours: one colour for 'A question I can now FULLY answer' and another for 'A question I still have OR a new question this lesson raised.' Learners write individually for 3 minutes, then in pairs they decide which of their 'can now answer' sticky notes best represents the whole lesson sequence's progress — they place one combined sticky note per pair on the DQB in the 'ANSWERED' zone. 'Still wondering' notes go in the 'STILL EXPLORING' zone. The teacher reads three 'still wondering' notes aloud — chosen to foreshadow Lesson 7's full resolution. The class votes by show of hands on which unanswered question is the MOST important to resolve. This question is circled on the DQB with a marker and labelled 'Resolve in Lesson 7.'	Distribute sticky notes and say: 'In the first lesson of this sequence, we put our big puzzles on this board. Look at it now. What has changed in your understanding?' Allow 3 minutes of silent individual writing. Enforce silence. After pairs place their notes, read three 'still wondering' notes aloud in a curious tone — do not express judgement. Say: 'These are excellent questions — they show you're thinking like physicists. Which one should we make sure to answer definitively in our next and final lesson?' Conduct the show-of-hands vote. Circle the winning question. Say: 'I am writing a commitment here: by the end of Lesson 7, this question will be fully answered using your own annotated models.' Point to the 'ANSWERED' zone and say: 'Look at how many questions have moved here since Lesson 1. That is the evidence of your learning — not just in your notebook, but on this wall.'	Metacognitive DQB Reflection — the dual-colour sticky note protocol forces learners to explicitly distinguish between consolidated understanding and residual uncertainty. This metacognitive act deepens encoding of resolved concepts and creates meaningful anticipation for Lesson 7. The public class vote democratises the learning agenda, increasing learner ownership of the final lesson's purpose.	Observable indicator: every learner produces at least one sticky note in each colour. The teacher scans 'still wondering' notes for misconceptions that indicate incomplete understanding (e.g., confusion between superconductivity and normal semiconductor behaviour, or conflation of doping with alloying). Any misconception appearing on three or more notes is addressed verbally by the teacher in 60 seconds before closing the phase, rather than letting it persist into Lesson 7.

Model Building Phase 📹 **VIDEO:** Threshold 8: The Modern Revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative annotated model — the diagram of the Kajiado solar panel and silicon atomic structure that has been revised across Lessons 1–5. Using a different colour pen (to show this lesson's additions), each learner individually adds: (1) a labelled arrow showing the direction of electron flow through the p-n junction when photons strike, (2) a temperature annotation showing why hot afternoon conditions reduce carrier mobility and lower output power, and (3) a 'technology connection' callout box in the margin linking their solar panel model to one of today's case studies (Turkana school panels, M-PESA microchip, or KNH MRI — learner's choice). Learners then write a two-sentence 'Model Caption' beneath their diagram that directly answers the driving question in their own words: 'The Kajiado solar panel produces less power on a hot afternoon because…, and silicon is special because…' The teacher collects or photographs 6–8 models (a purposive sample across ability groups) for review before Lesson 7.	Distribute or ask learners to open their cumulative model notebooks. Say: 'Every time we have learned something new in this sequence, you have added to this model. Today is the second-to-last addition. Use a NEW colour so we can see today's thinking clearly.' Allow 8 minutes of individual model revision — circulate and prompt quietly: 'Where on your diagram does the p-n junction sit? Can you label it? What changes when the temperature rises — draw that change.' After 8 minutes, say: 'Now write your two-sentence Model Caption at the bottom — answer both halves of our driving question. You have 3 minutes.' After writing, cold-call 3 learners to read their Model Caption aloud. For each, ask the class: 'Does this caption answer BOTH parts of the driving question? Thumbs signal.' If thumbs are sideways or down, ask: 'What one phrase would make it complete?' Allow 10 seconds WAIT TIME. Close by saying: 'In Lesson 7, you will use this model as the foundation for your final, fully-resolved explanation. Guard it — it is your intellectual work product for this entire sequence.'	Cumulative Annotated Model Revision with Colour-Coded Layering — by adding each lesson's conceptual gains in a distinct colour, learners create a visible record of their conceptual development. The model is not redrawn but grown, mirroring how scientific models evolve through iterative evidence-gathering. The two-sentence Model Caption requiring answers to both halves of the driving question forces integration rather than mere listing of facts.	Observable indicator: the updated model contains at minimum a labelled p-n junction, a directional electron-flow arrow, a temperature annotation (e.g., 'high T → more lattice vibration → lower carrier mobility → less current'), and a technology connection callout. The two-sentence Model Caption explicitly addresses both 'temperature reduces efficiency' and 'silicon is a semiconductor' in the learner's own words. The teacher uses the 6–8 photographed/collected models to identify: (a) learners who still show the current flowing in the wrong direction through the junction — these learners receive a targeted correction card at the start of Lesson 7; (b) learners whose captions are fully accurate and richly expressed — these learners are invited to share their models on the board at the start of Lesson 7 as anchor exemplars for the class.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal: (1) During the jigsaw phase, did each expert group demonstrate genuine depth of understanding, or did learners read directly from the sheet without internalising the reasoning? If reading verbatim was common, plan a 'closed-sheet explain' rehearsal before the jigsaw in future iterations. (2) When learners attempted to fill in the Kajiado Solar Panel row of the three-column board table, how complete and accurate were their explanations? Were both key ideas — temperature effect on carrier mobility AND the semiconductor nature of silicon — present, or did learners address only one? Use this to calibrate how much review to build into the opening of Lesson 7. (3) Review the 6–8 collected model diagrams tonight: are electron-flow arrows correctly directed from n-type to p-type through the external circuit? Is the temperature annotation mechanistically accurate (increased lattice vibrations reducing carrier mobility) rather than vague ('heat is bad')? Learners needing correction should receive a targeted written prompt returned at the start of Lesson 7. (4) Which learners were silent during whole-class discussion despite being cold-called — did WAIT TIME of 10 seconds feel sufficient, or should it be extended to 15 seconds for Lesson 7's more complex synthesis task? (5) Consider the equity dimension: did learners from all socioeconomic backgrounds engage with all three case studies, or did some learners disengage from the MRI/KNH case study as 'not relevant to my life'? If so, plan to explicitly connect the KNH MRI case to health access for all Kenyan families in the Lesson 7

opening.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We analysed three pieces of evidence: (1) a power-vs-temperature graph for a silicon solar panel in Lodwar showing power output falling as panel temperature rises from 15 °C to 45 °C; (2) a diagram of a p-n junction transistor switch inside a mobile processor showing how doping creates one-way electron flow that enables digital logic in M-PESA handsets; (3) a resistance-vs-temperature graph for a superconducting niobium-titanium alloy showing resistance dropping sharply to zero at ~9 K, as used in MRI magnets at Kenyatta National Hospital.
What did I learn?	We learned that the same family of semiconductor properties — band gaps, doping, p-n junctions, and temperature-dependent carrier behaviour — underpins three completely different Kenyan technologies. In solar panels, photons excite electrons across the band gap and the p-n junction drives them in one direction as useful current, but higher temperatures increase lattice vibrations that scatter carriers and reduce efficiency. In microchips, doped p-n junctions act as transistor switches that process digital signals at microscopic scales. In superconducting MRI magnets, cooling a metal alloy to near absolute zero collapses electrical resistance to exactly zero, enabling the powerful stable magnetic fields needed for medical imaging.
How does this explain the phenomenon?	We can now give a more complete answer to the driving question: the Kajiado solar panel produces less power on a hot afternoon because rising temperature increases lattice vibrations in the silicon crystal, which scatter charge carriers more frequently, reducing their mobility and therefore the current the p-n junction can deliver. Silicon is special because its band gap is neither too large (like an insulator) nor too small (like a metal) — it can be precisely engineered through doping to create junctions that convert light into directed electron flow. This understanding, developed by semiconductor physicists, now powers rural schools in Turkana, enables financial inclusion through M-PESA, and supports life-saving medical diagnosis at KNH.

LESSON 7 (80 minutes): Final Explanation: Semiconductors, Doping, and the Kajiado Solar Panel Mystery — Resolved	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To synthesise all evidence gathered across the unit into a coherent, scientifically accurate explanation of why silicon can act as both a conductor and an insulator, why the Kajiado solar panel produces less power in hot afternoon conditions, and how doping and p-n junctions underpin modern semiconductor technology.
Knowledge	Learners demonstrate understanding that: (a) conductors, semiconductors, and insulators differ in electrical properties due to their atomic and energy-band structures; (b) semiconductors such as silicon have conductivity that can be controlled through doping to form n-type and p-type materials; (c) a p-n junction converts light energy to electrical energy in a photovoltaic cell; (d) rising temperature increases resistance in semiconductors, reducing panel efficiency; (e) semiconductor technology has transformed modern technology and everyday life.
Skills	Learners are able to: (a) construct and present a final annotated scientific model that incorporates conductors, insulators, semiconductors, doping, and p-n junction behaviour; (b) compare their Lesson 1 initial model with their final model and articulate what changed and why; (c) use evidence from experiments, simulations, videos, posters, and summary tables to justify their explanations; (d) evaluate the completeness of the class Driving Question Board and close remaining questions.
Attitudes	Learners appreciate how the incremental accumulation of evidence — rather than a single observation — leads to robust scientific explanations, and they value the role of Kenyan and global semiconductor science in addressing real development challenges such as rural electrification in Kajiado.
Key Inquiry Question	Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun than on a cool morning — and what is so special about silicon that it can act as both a conductor and an insulator depending on the conditions?
Purpose in Storyline	This is the culminating lesson of the sub-strand. Learners have spent six lessons gathering evidence about atomic structure, energy bands, conductor/insulator differences, intrinsic semiconduction, doping, p-n junctions, and temperature effects. In Lesson 7 they bring all of that evidence together to write a full scientific explanation of the anchoring phenomenon, compare their naïve L1 model with their final model, complete the Driving Question Board, and connect semiconductor physics to modern Kenyan technology — closing the storyline that began with the puzzling Kajiado solar panel.
Safety Notes	No hazardous materials or electrical equipment are used in this lesson. If learners handle any electronic components (e.g., LED, solar cell) for the model-building demonstration, ensure hands are dry and components are handled gently. Remind learners not to look directly at a concentrated light source used to illuminate a solar cell.

B. LESSON OVERVIEW

Lesson 7 is the Final Explanation lesson for Sub-Strand 3.3. Learners begin by revisiting their Lesson 1 initial models of the Kajiado solar panel phenomenon and recording what they believed then. They then move through a structured evidence-synthesis activity — using their class posters, summary tables, and a curated YouTube clip on semiconductors and applications — to co-construct a complete scientific explanation. The teacher guides a whole-class 'Claim–Evidence–Reasoning' (CER) discussion anchored to the driving question, before learners individually revise and annotate their models. The lesson closes with a formal DQB completion ceremony, a teacher-led final explanation reading, peer assessment of final models, and a forward-looking connection to Kenyan solar-energy infrastructure and careers in electronics.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive their Lesson 1 initial model drawing (or a printed copy of the class consensus L1 model if individual copies are unavailable). Working in silence for 3 minutes, each learner annotates their L1 model in RED ink, marking every idea they now know to be incomplete, oversimplified, or wrong. They then write one sentence: 'In Lesson 1 I thought the solar panel worked less well in the heat because ___. I now think the real reason is ___.' Pairs share their sentences with each other before three are cold-called to share with the class. This activates prior knowledge, surfaces conceptual change, and sets up the synthesis that follows.	Distribute L1 models (or display class L1 model on screen). Say: 'Before we write our final explanation today, I want you to look at your very first idea — your Lesson 1 model. Take 3 minutes in silence. Use your RED pen and circle or cross out anything you now know is wrong or incomplete. Do not erase — we want to SEE the thinking change.' [Allow 3 minutes silent work.] Then say: 'Now complete this sentence in your notebook: In Lesson 1 I thought… I now think…' [Allow 2 minutes.] 'Turn and share with your partner.' [Allow 90 seconds.] Cold-call 3 learners: 'Amina, read us your sentence.' 'Kamau, what was the biggest change in your thinking?' 'Zawadi, what idea from Lesson 1 do you now know was incomplete — and what evidence changed your mind?' After each response, ask the class: 'Does anyone want to add to that?' [10-second wait time after each probe.] Record key 'I now know…' statements on the board in a two-column table: BEFORE | AFTER.	Strategy: CONCEPTUAL-CHANGE ELICITATION via annotated model comparison. This strategy makes the learner's own prior thinking visible as a target for revision, which research shows deepens retention of the correct scientific model. By requiring RED ink annotations rather than erasure, learners physically see the gap between their initial mental model and the evidence-based model — making the learning arc concrete and motivating.	Observable indicator: Each learner produces at least two RED annotations on the L1 model AND writes a complete BEFORE/AFTER sentence. Teacher circulates during silent work and looks for: (1) annotations that go beyond surface features (e.g., 'more sun = more electricity') to structural explanations (e.g., 'I didn't know about the p-n junction'); (2) learners who are still unable to articulate a change — these learners are noted for targeted support during Phase 3.
Observe Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure	Class posters and summary tables from Lessons 1–6 are displayed around the room (or projected in sequence if the room is small). Learners conduct a 10-minute structured GALLERY WALK in groups of three, completing a 'Evidence Harvest' table in their notebooks with four columns: LESSON | KEY EVIDENCE COLLECTED | WHAT IT EXPLAINS ABOUT THE PHENOMENON | QUESTIONS	Before the gallery walk, say: 'You have 10 minutes to walk around and visit every poster and summary table from our previous six lessons. You are detectives — your job is to harvest the best evidence. Fill in your Evidence Harvest table as you go. Focus especially on: What does this evidence tell us about why the Kajiado panel works less well in the heat?' [Set a visible 10-minute timer.] Circulate and prompt	Strategy: EVIDENCE HARVEST TABLE with GALLERY WALK. This strategy requires learners to actively categorise and connect evidence from multiple sources rather than passively receive a summary. By explicitly linking each piece of evidence to the anchoring phenomenon, learners practise the scientific practice of 'Using Mathematics and Computational Thinking' and 'Constructing Explanations' (NGSS SEP 6). The	Observable indicator: Each learner's Evidence Harvest table contains entries for at least 5 of the 6 previous lessons, with the 'What it explains about the phenomenon' column filled in (not left blank). Teacher checks 4–5 tables during the gallery walk. Red flag: any learner whose 'What it explains' column is empty — this signals the learner has been collecting facts without connecting them to the phenomenon.

[(Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	STILL OPEN. After the gallery walk, the teacher plays a 4-minute curated YouTube segment on semiconductors and their applications in solar cells and modern devices (Kenyan context: KPLC smart meters, M-KOPA solar home systems, mobile phone chips). Learners add any new evidence to their harvest table. Groups then have 2 minutes to flag their single most important piece of evidence on a sticky note and post it on the class EVIDENCE WALL under the driving question banner.	groups who are skimming: 'What did the doping experiment in Lesson 4 actually show you? Write that down.' After the gallery walk, say: 'Let's now watch a short video that connects what we've been studying to real applications in Kenya — things like M-KOPA solar panels and mobile phone processors.' [Play 4-minute clip.] Pause at 2 minutes and ask: 'What does this remind you of from our lessons?' [10-second wait. Cold-call 2 learners.] After the clip: 'Add anything new to your harvest table. Then with your group, pick your ONE most important piece of evidence — write it on a sticky note and post it under our driving question.' [Allow 3 minutes.]	sticky-note posting creates a shared, visible evidence wall that the class can reference during the CER discussion in Phase 3.	Intervention: teacher sits with that learner for 60 seconds and asks, 'How does THIS evidence help explain why the panel works less well at 35°C than at 18°C?'
Explain Phase 📹 **VIDEO:** The periodic table - classification of elements Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Effect of Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class co-constructs a Final Explanation of the anchoring phenomenon using the Claim–Evidence–Reasoning framework. The teacher writes the driving question on the board and guides a whole-class discussion to build three components: (1) CLAIM — a one-sentence answer to the driving question; (2) EVIDENCE — at least four pieces of specific, lesson-linked evidence supporting the claim; (3) REASONING — the scientific principles (energy band theory, doping, p-n junction, temperature effect on carrier recombination) that link the evidence to the claim. After the class co-constructs the CER on the board, each learner writes their own individual final explanation in their notebook using the CER structure. Learners also read and discuss the 'Introduction to Electronics – Final Explanation' text (provided on the ARES platform), annotating it with	Write on the board: 'DRIVING QUESTION: Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun than on a cool morning?' Say: 'We are now going to build our final scientific explanation together — using the Claim-Evidence-Reasoning framework. I need a CLAIM first — a one-sentence direct answer to the driving question. What is your claim?' [10-second wait. Cold-call 2 learners. Write the best claim on the board. Refine it as a class: 'Does everyone agree? Is it specific enough?'] Then: 'Now I need EVIDENCE. Look at your evidence harvest tables and the sticky notes on our wall. Call out specific, lesson-linked evidence — tell me the lesson and what you found.' [Build a numbered list of at least 4 pieces of evidence on the board. Probe: 'What lesson did that come from? How do we know that?'] Then: 'Now the	Strategy: CLAIM–EVIDENCE–REASONING (CER) CO-CONSTRUCTION followed by individual written explanation. CER is a structured argumentation framework (aligned with NGSS SEP 7: Engaging in Argument from Evidence) that forces learners to separate what they claim from what they can prove and from the scientific principles that do the explanatory work. Co-constructing first on the board lowers the cognitive load for individual writing, while the individual writing step ensures every learner produces their own accountable explanation. The Final Explanation text annotation step then bridges the co-constructed model to canonical scientific language, helping learners see their thinking reflected in formal text.	Observable indicator: Each learner's individual CER explanation includes: (1) a clear CLAIM sentence that directly answers the driving question; (2) at least FOUR specific evidence statements each linked to a lesson or activity; (3) REASONING that correctly uses at least THREE of the following terms in context: band gap, free electrons, semiconductor, doping, n-type, p-type, p-n junction, photovoltaic effect, temperature, carrier recombination. Teacher collects or photographs notebooks at the end of Phase 3. Learners who have fewer than three reasoning terms are given a 'Reasoning Scaffold' prompt card: 'The reason this evidence supports the claim is that silicon's _____ means that when _____, the result is _____.'

	connections to their own evidence. Finally, learners answer: 'How has the understanding and engineering of semiconductor materials transformed modern technology and everyday life in Kenya?' — referencing at least two Kenyan applications.	REASONING — the hard part. Why does that evidence support the claim? What is the scientific principle connecting them?' [Guide learners to articulate: free electrons in conductors vs. band gap in insulators; silicon's intermediate band gap as a semiconductor; doping creating n-type/p-type regions; p-n junction enabling photovoltaic effect; rising temperature increasing lattice vibrations and carrier recombination, reducing efficiency.] Say: 'Now write your OWN CER explanation in your notebook — in your own words. You have 8 minutes.' [Allow 8 minutes. Then say:] 'Now read the Final Explanation text on your ARES screen. Use a different colour pen to annotate: underline anything that matches your CER, and circle anything you didn't include that you should add.' [Allow 5 minutes.] Then ask: 'How has semiconductor science changed everyday life in Kenya specifically? Give me two examples.' [Cold-call 3 learners. Probe: 'Why does M-KOPA rely on semiconductor physics? What about the chips in a Safaricom mobile phone?']		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [The periodic table - classification of elements] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos]	The class returns to the Driving Question Board that has been built up across Lessons 1–6. Working in groups, learners review every question card on the DQB and vote on its status: GREEN = fully answered by our evidence; YELLOW = partially answered, we have some evidence; RED = still open, needs further study. Groups briefly discuss any RED cards and suggest what kind of investigation or source could answer them. The	Point to the DQB and say: 'This board has held all our questions since Lesson 1. Today we settle the score. In your groups, go through every card — give it a GREEN, YELLOW, or RED dot sticker based on how well our evidence has answered it. You have 4 minutes.' [Distribute dot stickers. Circulate, listening for reasoning.] After 4 minutes, bring the class together: 'Let's look at the RED cards — questions we	Strategy: DRIVING QUESTION BOARD TRAFFIC-LIGHT REVIEW. This metacognitive strategy asks learners to evaluate not just what they know but how well their evidence base answers specific questions — a practice aligned with NGSS SEP 8: Obtaining, Evaluating, and Communicating Information. The traffic-light coding makes epistemic status visible and communal, preventing learners	Observable indicator: The DQB contains dot stickers on every card, with the majority GREEN or YELLOW. Teacher notes any groups that mark the central driving question RED or YELLOW — this signals a gap in the unit's core explanation and triggers an immediate whole-class re-teaching moment. The three 30-second verbal answers are evaluated against a simple 3-point rubric (displayed on screen): 1 =

📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	class then conducts a 'DQB Ceremony': the teacher reads the original driving question aloud, and learners raise their hands if they feel they can now answer it confidently. Three learners are invited to give a 30-second verbal answer. The teacher closes by framing the RED (unanswered) questions as a bridge to future learning (e.g., transistors, integrated circuits, climate and space physics in upcoming strands).	couldn't fully answer. What would we need to do to answer them?' [Cold-call 2 groups. Accept all reasonable suggestions — e.g., 'We'd need to study transistors,' or 'We'd need to learn more about quantum mechanics.'] Then say: 'I'm going to read our driving question one more time. Raise your hand if you feel you can answer it now.' [Pause for a show of hands — aim for >80%.] 'Amina — give me your 30-second answer.' [Repeat for 2 more learners.] Then say: 'Every RED card on this board is a door to future physics. Transistors, integrated circuits, quantum computing — these are all built on the same silicon foundation we've been studying. You are standing at the beginning of that journey.' Write on the board: 'The RED questions on our DQB are NOT failures — they are the next frontier.'	from leaving the unit with the false impression that all science is settled. The 30-second verbal answer to the driving question is a low-stakes oral assessment that consolidates and publicly validates the unit's central learning.	mentions silicon/semiconductor; 2 = adds doping or p-n junction; 3 = correctly explains temperature effect on efficiency. Teacher records scores in the mark book as a formative grade.
Model Building Phase 📹 **VIDEO:** [The periodic table - classification of elements] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Effect of Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners produce their FINAL ANNOTATED MODEL — a diagram that tells the complete story of the Kajiado solar panel. The model must include: (a) atomic/energy-band diagram of silicon showing the band gap and how doping fills or empties the band; (b) a p-n junction diagram showing the depletion zone, direction of electron flow when light strikes, and the resulting current; (c) an arrow or annotation showing how rising temperature (35°C afternoon) increases lattice vibrations, increases carrier recombination, and reduces current output; (d) a comparison panel showing the cool morning (18°C) vs. hot afternoon (35°C) output. Learners then place their	Say: 'Your final task is to draw the model that explains everything — the complete story of the Kajiado solar panel, from silicon atoms all the way to the water pump switching on. Your model must include four things.' [Write on board: 1. Energy band diagram + doping. 2. p-n junction + photovoltaic effect. 3. Temperature effect at 35°C. 4. Cool morning vs hot afternoon comparison.] 'You have 12 minutes. Be detailed — label everything.' [Set timer. Circulate actively. Prompt: 'Where is the depletion zone in your diagram? Can you show me which way the electrons move when a photon hits? What happens to the lattice at 35°C — can I SEE that in your diagram?'] After 12 minutes:	Strategy: CUMULATIVE MODEL REVISION with SIDE-BY-SIDE L1 vs. FINAL COMPARISON and PEER GALLERY. This strategy enacts the NGSS SEP 2 (Developing and Using Models) at its highest level — learners do not just draw a diagram but use it as an explanatory and predictive tool. The side-by-side L1/final comparison makes conceptual change visible and assessable. The peer gallery with GLOW/GROW sticky notes builds metacognitive awareness and communication skills (Core Competency: Communication and Collaboration), while giving the teacher rapid formative data on the range of model quality across the class.	Observable indicator: The final model contains all four required components (energy band diagram, p-n junction, temperature effect, cool/hot comparison) and is labelled with correct scientific vocabulary. Teacher uses a 4-point holistic rubric: 4 = all four components present, correctly labelled, connected to the phenomenon; 3 = three components present, minor errors; 2 = two components, significant gaps; 1 = one component or major misconceptions present. The Model Evolution Statement is assessed for: (a) specificity of change described, (b) named evidence from a specific lesson, (c) intellectual honesty about remaining questions. Final models

	L1 model and their final model side by side and write a 'Model Evolution Statement': three sentences explaining what changed, what evidence drove each change, and what questions remain. Groups of four share their final models in a 2-minute gallery format — peers provide one 'glow' (something excellent) and one 'grow' (something that could be clearer) on a sticky note attached to the model.	'Now place your L1 model next to your final model. Write your three-sentence Model Evolution Statement — what changed, what evidence drove it, what is still open.' [Allow 4 minutes.] Then: 'Stand up — we are doing a 2-minute gallery. Walk around and read two other groups' final models. Leave a GLOW sticky note and a GROW sticky note on each one.' [Allow 5 minutes for gallery + sticky notes.] Bring the class together: 'What was the most impressive model you saw — and why?' [Cold-call 2 learners. Then say:] 'What you have drawn today is not just a school diagram. It is the same physics that engineers at companies like Azuri Technologies use to design solar panels for Kenyan homes. You understand semiconductor physics.'		are collected and scored as the summative assessment for the sub-strand.

D. TEACHER REFLECTION

After Lesson 7, reflect on the following questions before moving to Strand 4.0:

1. CONCEPTUAL CHANGE: When learners annotated their L1 models in RED, what were the most common misconceptions that persisted across the unit? (Common ones: 'more heat always helps electricity flow'; 'semiconductors are just weak conductors'; 'the solar panel works by heating up.') Did the six-lesson evidence arc successfully dislodge these? If not, what additional experience would have helped?

2. CER QUALITY: Were learners able to connect the REASONING component of their CER explanations to specific scientific principles (band gap, doping, p-n junction, temperature effect), or did they stop at describing observations? If reasoning was weak, consider adding a dedicated 'reasoning scaffold' lesson earlier in the unit in future iterations.

3. DQB COMPLETION: How many cards remained RED at the end of the DQB ceremony? If more than 20% of cards were still RED, this may indicate that the unit's evidence arc did not fully cover the scope of learner-generated questions — or that learners generated questions that appropriately go beyond Grade 10 content (e.g., quantum tunnelling, photon energy quantisation). Celebrate the RED cards as genuine intellectual frontier — do not treat them as failures.

4. MODEL QUALITY: Did final models show genuine integration of all four components, or were energy-band diagrams and p-n junction diagrams treated as separate, unconnected topics? If learners struggled to connect atomic structure to macroscopic panel behaviour, the 'scaling' link (atom → band → junction → panel → power output) may need to be made more explicit in Lessons 4–5 in future iterations.

5. KENYA CONTEXT: Did learners spontaneously reference M-KOPA, Azuri Technologies, KPLC smart meters, or Safaricom mobile chips during the CER discussion, or did these connections feel forced? Strong Kenyan contextualisation should feel natural by Lesson 7 — if it felt forced, consider embedding more Kenyan application case studies earlier (Lessons 2–3) rather than only in the final lesson.

6. EQUITY CHECK: Were all learners — including girls and learners from pastoralist communities in Kajiado — positioned as competent explainers during the DQB ceremony and model gallery? Note any patterns in cold-call responses and ensure that the summative model assessment values the quality of scientific reasoning over artistic drawing ability.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the class posters, summary tables, and video evidence from Lessons 1–6 — including the energy band diagrams, doping demonstrations, p-n junction models, and temperature-vs-efficiency data — all pointed to the same explanation. We also observed that our Lesson 1 models were incomplete: most of us thought 'more heat = more electricity,' but the evidence showed the opposite for semiconductors.
What did I learn?	We learned that silicon is a semiconductor because its band gap is small enough for electrons to jump from the valence band to the conduction band when given energy (light or heat), but large enough to block flow in darkness. Doping silicon with phosphorus (n-type) or boron (p-type) creates a p-n junction that drives electrons in one direction when light strikes — producing a current. At higher temperatures (like 35°C on a Kajiado afternoon), increased lattice vibrations cause more electron-hole recombination before electrons can do useful work, which reduces the panel's power output. On a cooler morning (18°C), recombination is lower and efficiency is higher — explaining the farmer's observation. Semiconductor technology has transformed modern life in Kenya: M-KOPA solar home systems, Safaricom mobile network chips, and KPLC smart meters all rely on the same physics we studied.
How does this explain the phenomenon?	We can now fully explain the anchoring phenomenon: The Kajiado solar panel produces less power in the hot afternoon not because there is less light, but because the higher temperature (35°C) increases lattice vibrations in the silicon crystal, raising the rate of electron-hole recombination in the p-n junction and reducing the number of electrons that complete the external circuit as useful current. On the cooler morning (18°C), recombination is lower, more photon-excited electrons cross the junction and flow through the circuit, and efficiency is higher. Silicon works as a solar panel material precisely because it is a semiconductor — its intermediate band gap allows light to excite electrons across the gap, while its p-n junction (created by doping) ensures those electrons flow in one direction as electricity. This is something neither a pure conductor (copper) nor a pure insulator (rubber) can do.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.